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Excess Movement in Option-Implied Beliefs

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ABSTRACT

We derive bounds on the rational variation in option-implied beliefs about market returns. While risk preferences distort option-implied (or *risk-neutral*) beliefs away from subjective beliefs, one can nonetheless bound risk-neutral belief movement under a general assumption on the stochastic discount factor. The resulting test requires no knowledge of the objective distribution and allows significantly more flexibility in discount rates than standard volatility tests. Implementing our test empirically using index options, we find that there is so much movement in risk-neutral beliefs that the bounds are routinely violated. Our results imply significant excess index-price volatility.

OPTION MARKETS ARE A RICH source of information about the perceived distribution of future stock returns. Like the overall market itself, option-implied beliefs about future returns are highly volatile over time. Testing whether they are too volatile relative to a rational benchmark, however, is a difficult task. First, subjective beliefs are not directly observable, as option prices also depend on unobservable risk and time preferences. Second, even if subjective beliefs

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were observable, we do not know the objectively correct conditional probabilities to use as a benchmark for comparison.

In this paper, we show that it is possible to place bounds on the rational variation in option prices, even without knowledge of the correct beliefs and with significant flexibility in preferences and discount rates. Implementing our bounds empirically, we find that there is so much price variation that the bounds are routinely violated. This implies that there must either be excess volatility in beliefs relative to rationality, or very large, high-frequency variation in the price of risk over small changes in the market index. In additional analysis, we find little evidence that rational variation in the price of risk is a key driver of our results, indicating that excess volatility in beliefs likely plays a role. Our findings imply significant excess volatility in the underlying price.

We focus our analysis on index options. As these options are effectively bets on the future value of the index, their prices can be used to measure a market-implied belief distribution over future index returns. Specifically, we use standard methods to transform option prices into so-called *risk-neutral (RN) beliefs*, which are a function of (i) the relative likelihood of different return states and (ii) the relative value of a marginal dollar in each state (which depends on risk preferences). Importantly, as RN beliefs are constructed using the *relative* prices of options with different strike prices but the same maturity, they strip out variation arising from time discounting and common unobservable shocks (e.g., to marginal utility or the quantity of risk) that have the same effect on options with different strikes. We can accordingly make informative statements about rational price variation with significantly weaker assumptions than in past volatility tests for the price of the underlying index.[1]

To introduce the logic of our test, we begin in a stripped-down setting and build up to an empirically realistic environment step by step. To isolate the first component of RN beliefs, assume for now that we can directly observe a person's subjective (or *physical*) beliefs $\pi_t(\theta = 1)$ over some binary outcome $\theta \in \{0, 1\}$. Suppose we observe π_t repeatedly oscillating from 0.90 to 0.10 and back as t progresses. While it is of course possible to rationalize this movement ex post by constructing a particular set of signal realizations from some data-generating process (DGP), this amount of movement appears intuitively rare for someone with rational expectations (RE). Formalizing this idea, Augenblick and Rabin (2021) note that when uncertainty is resolved by some period T, the expectation of the sum of squared changes in beliefs across all periods (*belief movement*, $\mathbb{E}\sum_{t=0}^{T-1}(\pi_{t+1} - \pi_t)^2$) must equal *initial uncertainty* ($\pi_0(1 - \pi_0)$) under RE, *regardless* of the DGP. Equivalently, expected *excess movement*—belief movement minus initial uncertainty, which we denote by $\mathbb{E}[X]$—must always be zero. Intuitively, belief movements imply learning, and rational learning implies a concomitant reduction in uncertainty (from the initial level $\pi_0(1 - \pi_0)$ to

[1] For example, Shiller (1981) famously documented excess volatility in stock prices relative to a measure of fundamental value, but his measure requires that discount rates over future cash flows remain constant over time. As discount rates appear to vary significantly, what to make of his results remains contested (Fama, 1991).

zero) on average. Therefore, if one observes a set of empirical belief streams for which excess movement is far from zero on average, the null of rationality can be statistically rejected at some confidence level. In other words, one can make statements about excess volatility in belief streams even without knowledge of the correct beliefs period by period.

The main theoretical contribution of this paper is to show how this logic—that movement in beliefs must correspond on average to reduction in uncertainty—can be used to restrict excess movement in risk-neutral beliefs (π_t^*) in the general case in which only risk-neutral rather than physical beliefs are observable. Our task becomes considerably more difficult in this case because risk-preference distortions allow RN excess movement $\mathbb{E}[X^*]$ to be nonzero, and its sign and magnitude depend on the degree of risk aversion and the precise form of the DGP. Put differently, nonzero excess movement in risk-neutral beliefs may in fact be consistent with rational updating. But even while allowing for meaningful degrees of freedom in risk preferences and in the form of the DGP, we show that the admissible $\mathbb{E}[X^*]$ can be bounded as a simple function of risk aversion over the terminal states (or, more generally, the slope of the stochastic discount factor, or SDF, across states).

The broad intuition carries over from the previous case. To illustrate, suppose that a person's valuation for a binary option that pays \$1 in state $\theta = 1$ at time T oscillates between \$0.90 and \$0.10, while her valuation for an option that pays \$1 in state $\theta = 0$ oscillates between \$0.10 and \$0.90. Risk-neutral beliefs π_t^* here are also moving back and forth between 0.90 and 0.10. But physical beliefs are not pinned down, as relative marginal utility (MU) in the two states is unknown. For example, the person might see the states as equally likely ex ante ($\pi_0 = 0.50$) but value the first option nine times higher because she values \$1 in state $\theta = 1$ nine times more. But if this marginal-utility ratio—which we call ϕ—is constant over time, then movement in the RN belief from 0.90 to 0.10 would mean that physical beliefs are oscillating between 0.50 and 0.01, which is still rare for a Bayesian. The same logic applies for any given value of ϕ as long as this parameter is relatively stable within a belief stream, an assumption we make to begin with and return to below.[2] Intuitively, extreme fluctuations in RN beliefs imply physical belief movements that must be rare regardless of the exact degree of risk aversion.

Our bounds for risk-neutral excess movement $\mathbb{E}[X^*]$ formalize this idea. The first bound that we derive is highly conservative: It holds under a particular "worst case" DGP that produces the highest possible movement for a given ϕ. This bound therefore requires no knowledge of the true statistical DGP for option prices, as it applies uniformly over the space of all possible DGPs. We then provide a set of tighter bounds that hold given additional information about the DGP. In both cases, the bounds are simple formulas that depend on the observable RN prior and are increasing in ϕ. Consequently, any observed

[2] Note that if ϕ can move arbitrarily, it becomes difficult to make statements about rational price variation. But as we discuss further below, arbitrary variation in ϕ is also difficult to make sense of under RE.

average excess movement in the data is informative as to the minimal risk-aversion value ϕ required for the bound to be satisfied.

How does this abstract setting apply to financial markets? In our empirical setting, the state θ depends on the value of the market index as of option expiration date T, or equivalently the return on the market over the life of the option. For example, suppose that the index value on date 0 is $S_0 = 100$. Define state $\theta = 1$ as occurring if S_T is equal to (or in a neighborhood close to) 100, or equivalently if the index gain R_T is close to 0%. Similarly, say $\theta = 0$ occurs if $S_T \simeq 105$. The RN belief π_t^* is then defined as the option-implied probability that $\theta = 1$ (i.e., $S_T \simeq 100$) conditional on $\theta \in \{0, 1\}$, which can be measured from the cross-section of options with different strike prices and expiration date T. This π_t^* plays the same role as in the abstract setting above, and its excess movement must satisfy our bounds. Importantly, the defined outcome θ stays fixed over the life of the option. This *fixed-event* setting is key for our results. It implies that the underlying subjective belief π_t must be a martingale under RE, and that learning must converge at date T: Either the index will be close to 100 or 105 (i.e., $R_T = 0\%$ or 5%), or neither state will be realized.[3]

In this asset-pricing context, the object that maps physical to RN beliefs is now $\phi = \frac{\mathbb{E}[M_T \mid R_T = 0\%]}{\mathbb{E}[M_T \mid R_T = 5\%]}$, where M_T is the SDF. This ratio of expected SDF realizations can be thought of as the market's local relative risk aversion, or price of risk, in a neighborhood of a 0% return realized from 0 to T. The assumption that ϕ is stable within a belief stream—which we maintain in deriving our main bounds—is therefore equivalent to assuming that the local price of risk in this neighborhood is constant. This is significantly weaker than the assumption of constant discount rates used, for example, by Shiller (1981): It allows for time-varying discount rates arising from multiple sources, including general changes in interest rates or the quantity of risk.[4] It is also met either exactly or approximately under many standard modeling frameworks.

That said, the assumption that the local price of risk ϕ is constant within a belief stream is a knife-edge restriction that is unlikely to hold exactly, and it is also violated in some leading option pricing models with priced stochastic volatility. We therefore also consider how our bounds change given variation in ϕ. We find, both theoretically and in a set of simulations, that rational variation in ϕ has a limited impact on RN belief movement. This may seem counterintuitive: One might imagine that if ϕ oscillates between 3.1 and 2.9, for example, then RN belief movement can be unbounded even with no movement in physical beliefs. While this is true, it overlooks the fact that these oscilla-

[3] By contrast, in many common settings, the fundamental value is a moving target. For example, the time-t subjective distribution over the *fixed-horizon* outcome $S_{t+\tau}$ may have lower variance than the $t+1$ distribution over $S_{t+1+\tau}$. See Nordhaus (1987) for discussion of fixed-event versus fixed-horizon tests, and see Kelly et al. (2026) for a recent example.

[4] For example, suppose that the second-period option prices in the example above were \$0.05 and \$0.45 (for states 1 and 0) rather than \$0.10 and \$0.90, due to a shift in the time-t value of money. This shift induces no change in risk-neutral beliefs relative to the example, because the relative prices are unchanged. Meanwhile, such a change would be problematic for any single-asset volatility test assuming constant discount rates.

tions in ϕ are *also* inconsistent with rational updating: They imply predictable mean reversion in the expected marginal value of \$1 in a fixed terminal state. A bound violation thus suggests that either physical beliefs or risk prices are excessively volatile. We return to the question of their relative importance in our empirical analysis.

After providing and discussing our theoretical results, we next take our bounds to the data. We obtain S&P 500 index option prices from OptionMetrics, and we use standard methods to infer the risk-neutral distribution over index returns for each option expiration date in the sample. We then translate each full distribution into a set of binary RN beliefs $\pi_t^*(R_T = \theta_j \,|\, R_T \in \{\theta_j, \theta_{j+1}\})$; these correspond to the RN probability that the index return will be close to θ_j, conditional on being either θ_j or θ_{j+1}. The return states correspond to 5 percentage point (pp) ranges for the S&P return, matching the example above. We then implement our theoretical bounds, which allow us to infer the minimal local risk-aversion value ϕ (at each point in the return distribution) needed to rationalize the observed variation in RN beliefs over the index return.[5]

Our main empirical finding is that there is so much excess movement in RN beliefs that extremely high risk aversion is needed to rationalize the data under our maintained assumptions. In many cases, there is in fact no value of ϕ under which the tight version of the bound is met, and the conservative version of the bound generally implies implausibly large values for ϕ. This suggests that many leading frameworks capable of explaining variation in underlying asset prices have difficulty rationalizing the large degree of observed variation in RN beliefs.

To better understand the implications of our results, we conduct three further analyses. First, we simulate two benchmark option pricing models to see whether they generate RN belief movement close to our empirical estimates. While they produce a moderate degree of excess movement, it is far below what we observe in the data. So despite the models' ability to explain many empirical regularities in option markets, they have difficulty generating the strong and predictable mean reversion in relative option prices that we find. We are of course not the first to provide evidence that well-known models are inconsistent with various moments, but our results offer progress in understanding what specifically can be deduced from the observed volatility in asset prices.

Second, to begin to disentangle possible explanations for our results—predictable mean reversion in either (i) physical beliefs or (ii) the price of risk, or alternatively (iii) option-market frictions—we conduct a set of reduced-form regressions for observed excess movement. We find a strong relationship between excess movement and proxies for overreaction in expectations, and little explanatory power from proxies for risk-price variability or market frictions.

[5] Given that we conduct our estimation using variation in index option prices, we must also account for the effect of nonfundamental or microstructure noise. To do so, we estimate the variance of the noise component of observed RN beliefs using a set of intraday option prices and the microstructure noise variance estimator proposed by Li and Linton (2022). We then construct an empirical noise correction, removing the effect of noise from X^* before we conduct our estimation. Our tests are therefore constructed to be robust to idiosyncrasies specific to the option market.

This suggests that overreaction by the marginal investor is likely a necessary feature in explaining our results.

Finally, to conclude our analysis, we conduct a rough quantification of how our results on RN beliefs translate to excess volatility in the underlying index. We do so by constructing an alternative index price from the RN distribution in which we exclude tail return states, and we show how excess movement in this object helps discipline excess volatility at the index level. Using this object, we estimate that the variance of the S&P 500 is about 20% to 30% higher than it would be under a martingale-belief benchmark. Our results on RN beliefs therefore imply meaningful excess volatility in the index itself.

Relation to previous literature: In addition to Shiller (1981), we follow, among others, LeRoy and Porter (1981), West (1988), and Stein (1989) in testing for excess volatility in asset prices relative to measures of fundamental value. Marsh and Merton (1986) emphasize nonstationarity in accounting for apparent excess volatility; much of the subsequent literature has emphasized time variation in discount rates (Cochrane, 2011). By contrast, our test allows for variation in discount rates and does not require any measure of fundamental value.

While our framework requires significantly weaker theoretical assumptions than classic volatility tests, one cost of this generality is that we rely on option prices rather than measure the behavior of the underlying index directly. In doing so, we follow a long line of work using options for information about more general expectations or quantities of interest.[6] Our work particularly complements Giglio and Kelly (2018), who document excess volatility in long-maturity claims on volatility, inflation, commodities, and interest rates. They achieve identification by parameterizing the DGP for cash flows on the underlying, whereas we restrict the SDF. Their parameterization—an affine model under the RN measure—applies well to the term-structure-like claims they consider, but not to claims on the equity index itself, to which our framework does apply.[7]

Our results also complement evidence obtained from survey data, as, for example, in Greenwood and Shleifer (2014) and De la O and Myers (2021). Another set of related literature endeavors to measure physical probabilities indirectly from asset prices. For example, Aït-Sahalia, Wang, and Yared (2001) test whether option-implied return distributions are well calibrated, and Polkovnichenko and Zhao (2013) consider the probability weighting functions consistent with option-price data given an assumption on the form of the weighting function. Our approach differs from this and related work in that we need not measure physical beliefs at all or know the true DGP for returns to conduct our tests. This semi-parametric approach loosely ties our theoretical contribution to a line of work including, among many others, Hansen and

[6] A recent sample includes Backus, Chernov, and Martin (2011) regarding disaster risk, Martin (2017) and Beason and Schreindorfer (2022) on the equity premium, and Haddad, Moreira, and Muir (2025) on the impact of policy promises.

[7] See also Gandhi, Gormsen, and Lazarus (2026) for recent evidence on the term structure of the equity premium.

Jagannathan (1991) and Alvarez and Jermann (2005), who consider identification of structural parameters from different moments of the observable data than those we consider here.

Organization: Section I introduces our theory in a two-state setting, which allows for clear derivations and intuition for our main results. Section II extends these results to a general asset-pricing setting. Section III discusses our identifying assumptions and provides robustness results. We implement our bounds empirically in Section IV, and in Section V we discuss the interpretation of our results. Section VI concludes. Proofs of our main results are provided in the Appendix, and an Internet Appendix contains the remaining proofs and additional technical material.[8]

I. Theoretical Framework: Introduction in a Simple Setting

To introduce our framework, we first examine risk-neutral belief movement in a simple setting with a single person and two terminal states. This setting helps clarify (i) the economics underlying restrictions on belief movement under RE when the individual is risk-neutral, (ii) how risk aversion complicates this analysis, and (iii) how we can nonetheless bound RN belief movement. This step-by-step discussion sets the stage for our generalized framework in Section II.

A. Setup and Initial Results

Time is discrete and indexed by $t = 0, 1, 2, \ldots, T$. At the beginning of each period, a person observes a signal $s_t \in S$ regarding two mutually exclusive and exhaustive states $\theta \in \{0, 1\}$. The DGP is general: Signals are drawn from the discrete signal distribution $DGP(s_t \mid \theta, H_{t-1})$, where H_t is the history of signal realizations through t. Define $\mathbb{P}(H_T)$ to be the probability of observing history H_T induced by the DGP, and denote the expectation under $\mathbb{P}$ by $\mathbb{E}[\cdot] \equiv \mathbb{E}^{\mathbb{P}}[\cdot]$. The person's (*physical* or *subjective*) *belief* in state 1 (versus state 0) at time t given the DGP and history H_t is denoted by $\pi_t(H_t)$. The *belief stream* $\boldsymbol{\pi}(H_t) = [\pi_0, \pi_1(s_1), \pi_2(\{s_1, s_2\}), \ldots]$ is the collection of beliefs given H_t. We often suppress the dependence of these objects on H_t to simplify notation. Given our empirical setting, we focus on *resolving* streams in which the person achieves certainty about the true state by period T with probability one: $\pi_T(H_T) = \theta \in \{0, 1\}$ for all H_T.

Throughout the paper, we maintain the assumption that the person's beliefs over the terminal state satisfy RE.

ASSUMPTION 1 (RE): *Beliefs satisfy* $\pi_t(H_t) = \mathbb{E}[\theta \mid H_t]$ *for any* H_t.

Assumption 1 states that the agent's beliefs coincide period by period with the true conditional probability of realizing state $\theta = 1$; this assumption is

[8] The Internet Appendix is available in the online version of the article on *The Journal of Finance* website.

satisfied by a person with a correct prior who updates using Bayes' rule according to the true DGP. The assumption is stronger than necessary for our main results: We could instead assume the weaker martingale restriction that $\pi_t = \mathbb{E}[\pi_{t+1} \mid \pi_t]$, which is implied by Assumption 1. Assuming RE directly, though, helps streamline exposition.

Given our asset-pricing setting, we assume that the DGP and the individual's physical beliefs cannot be observed directly. Instead, we assume that the econometrician can observe, period by period, the person's willingness to pay for an Arrow-Debreu security that pays \$1 (one unit of the numeraire consumption good) in period T if state θ is realized. Denote this valuation by $q_t(\theta|H_t)$ for each $\theta \in \{0, 1\}$. An object analogous to $q_t(\theta|H_t)$ is empirically observable using options data for suitably defined states, but we postpone this additional formalism to Section II.

We start by considering the simple case in which the person values consumption in all periods and states equally (i.e., she is risk-neutral and does not discount future consumption). In this case, beliefs are in fact directly inferable from asset values: $q_t(1|H_t) = \pi_t(H_t)$ and $q_t(0|H_t) = 1 - \pi_t(H_t)$. Testable restrictions on prices are thus equivalent in this case to restrictions on physical beliefs.

We keep track of the following objects related to the physical belief stream, and we discuss below how these objects are restricted under RE. First, total *belief movement* of $\boldsymbol{\pi}$ is defined as the sum of squared changes (or quadratic variation) in beliefs across all periods:

$$\mathsf{m}(\boldsymbol{\pi}) \equiv \sum\nolimits_{t=0}^{T-1} (\pi_{t+1} - \pi_t)^2. \tag{1}$$

Second, *initial uncertainty* of $\boldsymbol{\pi}$ is defined as the variance of the Bernoulli random variable $\mathbb{1}\{\theta = 1\}$ as of $t = 0$:

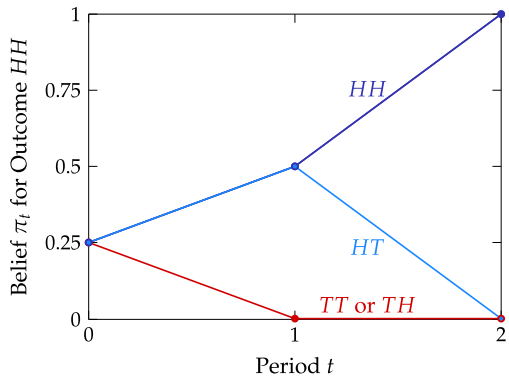
$$\mathsf{u}_0(\boldsymbol{\pi}) \equiv (1 - \pi_0)\pi_0. \tag{2}$$

Given that we focus on resolving belief streams for which $\pi_T \in \{0, 1\}$, final uncertainty is always zero ($\mathsf{u}_T = (1 - \pi_T)\pi_T = 0$). Initial uncertainty u_0 is therefore equal to the total amount of uncertainty reduction for $\boldsymbol{\pi}$, $\mathsf{u}_0 - \mathsf{u}_T$, which is helpful in interpreting some of the following results.

Our main variable of interest is the difference between movement and initial uncertainty, which—for reasons that will become clear—we call *excess movement*:

$$X(\boldsymbol{\pi}) \equiv \mathsf{m}(\boldsymbol{\pi}) - \mathsf{u}_0(\boldsymbol{\pi}). \tag{3}$$

Belief movement and initial uncertainty are related under RE according to Lemma 1, which restates a main result used by Augenblick and Rabin (2021) that has also appeared in other forms in past literature (e.g., Kyle, 1985, p. 1329; Barndorff-Nielsen and Shephard, 2001, p. 180; Collin-Dufresne and Fos, 2016, footnote 5).



STATISTICS

Signals (s_1, s_2)	Frequency $\mathbb{P}(s_1, s_2)$	Movement m
HH	0.25	0.3125
HT	0.25	0.3125
TT or TH	0.5	0.0625

$\mathbb{E}[\mathsf{m}] = 0.1875 = \mathsf{u}_0$
$\Longrightarrow \mathbb{E}[X] = 0$

Figure 1. Physical beliefs and excess movement: Two-period example. Signals (s_1, s_2) are generated by sequential fair coin flips. State $\theta = 1$ is realized if two heads occur: $(s_1, s_2) = (H, H)$, labeled HH; otherwise $\theta = 0$. Each path in the figure corresponds to a possible belief stream over $\theta = 1$ for a person with RE. In the table, m, u_0, and X are as defined in (1) to (3), and $\mathbb{E}[\mathsf{m}] = 0.25 \times 0.3125 \times 2 + 0.5 \times 0.0625$. (Color figure can be viewed at wileyonlinelibrary.com)

LEMMA 1 (Augenblick and Rabin, 2021): *Under Assumption 1, for any DGP, expected total belief movement must equal initial uncertainty. Expected excess movement in beliefs must therefore be zero:*

$$\mathbb{E}[X] = 0.$$

Lemma 1 is a straightforward implication of the assumption of martingale beliefs.[9] The result formalizes a notion of the "correct" amount of belief volatility under RE, and it motivates our use of the term *excess movement* for X. Given a set of observed belief streams, one can straightforwardly calculate the sample average of the empirical excess movement statistic and test if it differs from zero. The restriction reflects the intuition that if the person's beliefs are moving, this movement must on average correspond to learning about the true terminal state (in the sense that uncertainty is resolved from its initial value to 0). Rewriting $\mathbb{E}[X] = 0$ as $\mathbb{E}[\sum_{t=0}^{T-1}(1 - 2\pi_t)(\pi_{t+1} - \pi_t)] = 0$ (see footnote 9), it is apparent that expected belief movements toward 0.5 (the point of highest uncertainty) lead to a positive $\mathbb{E}[X]$ statistic, and vice versa. So it could be the case that $\mathbb{E}[\pi_{t+1} - \pi_t] = 0$ unconditionally, but a test based on the lemma would still reject the null of RE if, for instance, low values of π_t ($\pi_t < 0.5$) tend to be revised upward ($\pi_{t+1} - \pi_t > 0$) and high values tend to be revised downward.

To illustrate our setting and the lemma above, Figure 1 plots belief streams under a two-period DGP. Two fair coins are flipped sequentially at $t = 1$ and 2, generating two possible signals (H or T) at each of these dates. If two heads occur (HH), then state $\theta = 1$ is realized; otherwise, $\theta = 0$ is realized. As the

[9] To see this, rewrite X as $\sum_{t=0}^{T-1}(2\pi_t - 1)(\pi_t - \pi_{t+1})$. Using the law of iterated expectations on each term in the sum, $\mathbb{E}[(2\pi_t - 1)(\pi_t - \pi_{t+1})] = \mathbb{E}[(2\pi_t - 1)(\pi_t - \mathbb{E}[\pi_{t+1}|\pi_t])]$, which must be zero under the martingale assumption.

figure shows, the person's prior π_0 is equal to 0.25 under RE (equal to the probability of two heads). If the first coin flip is tails (shown in the red path), then $\pi_1 = \pi_2 = 0$; if heads then tails are flipped, then $\pi_1 = 0.5$, $\pi_2 = 0$ (light blue); and if two heads are flipped, then $\pi_1 = 0.5$, $\pi_2 = 1$ (dark blue). The accompanying table shows belief movement $\mathsf{m} = (\pi_1 - \pi_0)^2 + (\pi_2 - \pi_1)^2$ for each possible stream. Weighting the paths by their frequencies, expected belief movement is $\mathbb{E}[\mathsf{m}] = 0.1875$. This is exactly equal to initial uncertainty $\mathsf{u}_0 = (1 - 0.25) \times 0.25 = 0.1875$, illustrating the restriction that $\mathbb{E}[X] = 0$. This applies beyond this two-period example: The lemma implies that *any* DGP for which $\pi_0 = 0.25$ would generate expected movement of 0.1875 under RE.

B. Risk-Neutral Beliefs: Setup and Identification Challenge

Lemma 1 shows that one can make statements about excess belief movement under RE when beliefs are directly observable or inferable from asset prices. But identifying excess movement in asset valuations becomes significantly more complicated when the person is risk-averse, as valuations no longer correspond directly to beliefs in this more general case. To see this, assume now that the agent has time-separable utility, with concave period utility function $U(C_t)$, and that she exponentially discounts future consumption with discount factor β. Assume that the state θ determines period-T consumption $C_{T,\theta}$. Valuations for the two Arrow-Debreu securities at time t are now

$$q_t(1|H_t) = \frac{\beta^{T-t}U'(C_{T,1})}{U'(C_t)}\pi_t, \qquad q_t(0|H_t) = \frac{\beta^{T-t}U'(C_{T,0})}{U'(C_t)}(1-\pi_t). \tag{4}$$

Valuations therefore no longer directly reveal beliefs. Instead, they are distorted by the relative value of a marginal dollar in state θ in period T versus one in period t. As $q_t(1|H_t)$ and $q_t(0|H_t)$ are similarly distorted by β^{T-t} and $U'(C_t)$, it is useful to focus on their relative valuations. This logic leads to the consideration of *risk-neutral (RN)* beliefs,

$$\pi_t^*(H_t) \equiv \frac{q_t(1|H_t)}{q_t(0|H_t) + q_t(1|H_t)} = \frac{U'(C_{T,1})}{\mathbb{E}_t[U'(C_T)]}\pi_t(H_t) = \frac{\phi\pi_t(H_t)}{1 + (\phi - 1)\pi_t(H_t)}, \tag{5}$$

$$\text{where} \quad \phi \equiv \frac{U'(C_{T,1})}{U'(C_{T,0})}. \tag{6}$$

The definition in (5) follows the usual convention for RN beliefs, and the remaining expressions follow from (4). Like π_t, the RN belief π_t^* corresponds to state $\theta = 1$. The RN belief for $\theta = 0$ can be similarly defined as $\frac{q_t(0|H_t)}{q_t(0|H_t)+q_t(1|H_t)} = 1 - \pi_t^*$, so the two states' RN beliefs are positive and sum to one. Define the RN belief stream by $\boldsymbol{\pi}^*(H_t)$, RN belief movement by $\mathsf{m}^*(\boldsymbol{\pi}^*)$, RN initial uncertainty by $\mathsf{u}_0^*(\boldsymbol{\pi}^*)$, and RN excess movement by $X^*(\boldsymbol{\pi}^*)$ as in Section I.A, but with RN beliefs π_t^* in place of π_t.

Risk-neutral beliefs are so named because they can be interpreted as the subjective beliefs for a fictitious risk-neutral agent. In general, they rep-

resent a pseudo-belief distribution, reflecting a combination of the person's physical beliefs π_t and risk preferences as indexed by ϕ. This object ϕ will be particularly important in our environment. It represents the (constant) marginal rate of substitution across primitive states, as can be seen in (6). Up to a scaling constant, it also serves as an index of relative risk aversion (RRA): Conducting a Taylor expansion of $U'(C_{T,0})$ around $C_{T,1}$ gives $U'(C_{T,0}) = U'(C_{T,1}) + U''(C_{T,1})(C_{T,0} - C_{T,1}) + \mathcal{O}\big((C_{T,0} - C_{T,1})^2\big)$, and thus, to first order,

$$\gamma(C_{T,1}) \equiv -\frac{C_{T,1}U''(C_{T,1})}{U'(C_{T,1})} = \frac{\phi - 1}{(C_{T,0} - C_{T,1})/C_{T,1}}. \tag{7}$$

Relative risk aversion γ thus depends on the ratio of marginal utilities across states ϕ relative to the percent consumption gap across states. Finally, in asset-pricing terms, ϕ is equivalent to the ratio of SDF realizations $M_{t,T}(\theta)$ across the two states:

$$\phi = \frac{M_{t,T}(1)}{M_{t,T}(0)}, \quad \text{where} \quad M_{t,T}(1) \equiv \frac{q_t(1|H_t)}{\pi_t(H_t)}, \quad M_{t,T}(0) \equiv \frac{q_t(0|H_t)}{1 - \pi_t(H_t)}.$$

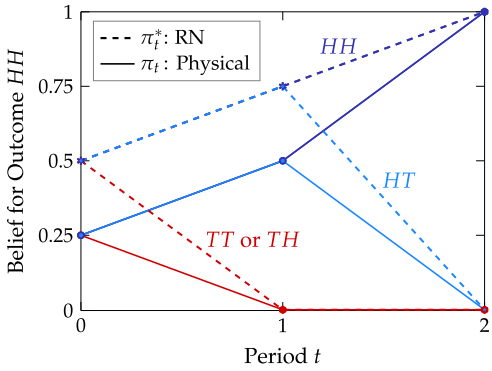
This SDF-based representation of ϕ will be discussed in detail in the general setting in Section II.

To complete the setup for the current consumption-based setting, we assume that consumption in state 1 is weakly less than in state 0, $C_{T,1} \leqslant C_{T,0}$. This is without loss of generality for now, as the states can be relabeled arbitrarily. With concave utility, this labeling implies that $U'(C_{T,1}) \geqslant U'(C_{T,0})$ and thus $\phi \geqslant 1$. Given $\phi \geqslant 1$, the RN belief π_t^* in general exceeds the subjective belief π_t: The person is willing to pay relatively more for a bad-state consumption claim given her high marginal utility in that state, upwardly biasing the bad-state RN belief relative to π_t.

We wish to make statements similar to Lemma 1, but for observable RN beliefs rather than physical beliefs. Under a risk-neutral expectation $\mathbb{E}^*[\cdot]$ defined such that $\pi_t^*(H_t) = \mathbb{E}^*[\pi_{t+1}^*(H_{t+1}) \,|\, H_t]$, one could in fact apply Lemma 1 directly, as $\mathbb{E}^*[X^*] = 0$. But the frequency of observed RN belief streams is determined by the physical measure rather than the RN measure; that is, we can only estimate $\mathbb{E}[X^*]$ rather than $\mathbb{E}^*[X^*]$. Moreover, even under the maintained assumption that physical beliefs π_t satisfy RE, the distortion in π_t^* relative to π_t can cause RN movement to differ from RN initial uncertainty on average, so $\mathbb{E}[X^*] \neq 0$. This is the fundamental identification challenge.

To illustrate this issue, Figure 2 returns to the coin-flip example from Figure 1. In addition to physical beliefs, this figure now shows RN belief streams with $\phi = 3$.[10] From (5), the physical prior $\pi_0 = 0.25$ corresponds to an RN prior of $\pi_0^* = 0.5$. Intuitively, the person perceives state $\theta = 0$ as three times as likely as $\theta = 1$ (HH), but values a marginal dollar in $\theta = 1$ three times more than in $\theta = 0$, so $q_0(0) = q_0(1)$ and $\pi_0^* = 0.5$. The same calculations are then

[10] Under risk neutrality, $\phi = 1$. This was implicitly assumed to be the case in Figure 1.



STATISTICS

Signals (s_1, s_2)	Frequency $\mathbb{P}(s_1, s_2)$	RN Movement m^*
HH	0.25	0.125
HT	0.25	0.625
TT or TH	0.5	0.25

$$\mathbb{E}[\mathsf{m}^*] = 0.3125 > 0.25 = \mathsf{u}_0^*$$
$$\Longrightarrow \mathbb{E}[X^*] = 0.0625 > 0$$

Figure 2. RN beliefs and excess movement: Two-period example. See Figure 1 for a description of signals and physical beliefs. Observed RN beliefs are calculated using (5), with the assumption that $\phi = 3$. In the table, m^*, u_0^*, and X^* are calculated as in (1) to (3), with RN beliefs π_t^* in place of π_t. (Color figure can be viewed at wileyonlinelibrary.com)

applied to obtain π_1^* and π_2^* for each stream shown in the figure. The accompanying table shows RN movement m^* for each stream. Comparing this table to the table in Figure 1, it is clear that $\mathsf{m}^* > \mathsf{m}$ for streams in which the RN belief has large downward revisions (HT, TT, and TH): π_t^* is biased upward relative to π_t, so these large downward revisions generate more RN movement than physical movement. These streams cause RN excess movement to be positive on average: $\mathbb{E}[X^*] = 0.0625 > 0$.

As this example illustrates, even with rational physical beliefs, one can observe what appears to be excess movement in RN beliefs implied by valuations. So if we naively test for RE using Lemma 1 on observed RN beliefs, we may spuriously conclude that beliefs are excessively volatile.

C. Risk-Neutral Beliefs: Results

The fact that there can be excess movement in RN beliefs even under RE would seem to pose an intractable challenge. But this turns out not to be the end of the story. RN beliefs are not arbitrarily distorted relative to physical beliefs: As in (5), they are linked through the single unobserved parameter ϕ. And regardless of the value of ϕ, RN beliefs must lie between zero and one by definition. These insights will allow us to bound $\mathbb{E}[X^*]$ over all possible DGPs for a given value of ϕ, and then over all possible values of ϕ. Because our main results are most straightforwardly stated in the current section's two-state setting, we state them here before discussing how they generalize in Section II.

C.1. Main Results

We first define two useful additional objects. First, we invert (5) to solve for π_t as a function of π_t^* and ϕ, the solution to which we denote by $\pi_t(\pi_t^*, \phi)$:

$$\pi_t(\pi_t^*, \phi) = \frac{\pi_t^*}{\phi + (1-\phi)\pi_t^*}. \tag{8}$$

Second, it will be helpful to define the difference in conditional expected X^* across states as

$$\triangle \equiv \mathbb{E}[X^*|\theta = 0] - \mathbb{E}[X^*|\theta = 1]. \tag{9}$$

Given these definitions, we can now provide a number of expressions and bounds for $\mathbb{E}[X^*]$. We assume throughout that Assumption 1 holds.

PROPOSITION 1: *For any DGP,*

$$\begin{aligned}\mathbb{E}[X^*] &= (\pi_0^* - \pi_0)\triangle \\ &= \left(\pi_0^* - \frac{\pi_0^*}{\phi + (1-\phi)\pi_0^*}\right)\big(\mathbb{E}[X^*|\theta = 0] - \mathbb{E}[X^*|\theta = 1]\big).\end{aligned}$$

PROPOSITION 2: *For any DGP and any value for* $\triangle$*,*

$$\begin{aligned}\mathbb{E}[X^*] &\leqslant (\pi_0^* - \pi_0)\pi_0^* \\ &\leqslant \left(1 - \frac{1}{\phi + (1-\phi)\pi_0^*}\right)\pi_0^{*2}.\end{aligned} \tag{10}$$

Proofs for this section's main results are provided in the Appendix. Proposition 1 starts from the fact that $\mathbb{E}^*[X^*] = 0$. We then connect $\mathbb{E}^*[X^*]$ to $\mathbb{E}[X^*]$. The key step is to show that *conditional* expectations of X^* under both measures are equal, $\mathbb{E}^*[X^*|\theta] = \mathbb{E}[X^*|\theta]$ for $\theta = 0, 1$, which leads to the stated result. Note that if $\phi = 1$, then $\pi_0^* = \pi_0$ and therefore $\mathbb{E}[X^*] = 0$, as in Lemma 1. As ϕ rises, π_0 drops further below π_0^*, and $\mathbb{E}[X^*]$ departs from zero: The greater is risk aversion, the more one can observe excess RN movement differ from zero on average under RE. The sign and magnitude of this deviation depend on $\triangle$, as we explore in more detail later in this section.

Across all DGPs, $\triangle$ is bounded above by π_0^*, from which Proposition 2 follows. The version of the bound in (10) is one of our main results. It gives a bound for $\mathbb{E}[X^*]$ as a function of π_0^* and ϕ, regardless of $\triangle$. Under risk neutrality ($\phi = 1$), this upper bound again becomes zero. But the bound is otherwise positive, and the *admissible excess movement* in RN beliefs given by the right side of the inequality increases monotonically in ϕ. This result thus formalizes a more general notion of the admissible amount of belief volatility under rationality, this time as an increasing function of risk aversion across the two states.

Intuitively, movement in RN beliefs must still correspond on average to the agent learning something about the true terminal state, but the bias in RN

relative to subjective beliefs induced by risk aversion allows for positive excess movement in those observed beliefs under RE. This is reflected in the first term in the bound, $(\pi_0^* - \pi_0)$, as this difference increases in ϕ. For the second term (π_0^*), higher RN priors yield more "room" for downward RN belief movement. This increases admissible excess movement along the lines of the example considered in Figure 2. The bound in Proposition 2 is conservative, as it holds under a "worst-case" DGP with an extreme value for $\triangle$.

Because of the monotonicity of the bound in ϕ, the inequality can also be read as providing a *lower* bound for the unobserved structural parameter ϕ as a function of observables. The observed excess movement in RN beliefs thus provides information on the minimal value of risk aversion necessary for the data to be consistent with RE. But more still can be said: Taking $\phi \to \infty$ in (10) generates the following maximally conservative bound for $\mathbb{E}[X^*]$, which applies for *any* ϕ.

COROLLARY 1: *For any DGP and any values* $\triangle$ *and* ϕ*,*

$$\mathbb{E}[X^*] \leqslant \pi_0^{*2}.$$

Corollary 1 exploits the fact that RN beliefs are bounded between zero and one by construction: There is a limit to how far RN beliefs can be distorted relative to subjective beliefs, so the bound is well defined even for arbitrarily high risk aversion.[11] The corollary thus states that price movements for which $\mathbb{E}[X^*] > \pi_0^{*2}$ simply cannot be rationalized under RE given constant ϕ. Despite its maximal conservatism, this bound still imposes a meaningful limit on admissible RN excess movement, especially for low π_0^*. For example, with $\pi_0^* = 0.2$, RN excess movement is at most 0.04, regardless of ϕ or the DGP.

Together, Proposition 2 and Corollary 1 characterize the maximal admissible excess movement in RN beliefs as a function of ϕ for any RN prior. Figure 3 graphically illustrates these bounds. Starting from the bottom of the chart, the thick purple line shows the bound for $\phi = 1$: In this case, $\mathbb{E}[X^*] = \mathbb{E}[X] = 0$ regardless of the prior or DGP, from Lemma 1. The thin dashed gray lines correspond to arbitrarily selected DGPs in the case of $\phi = 3$. While there can be positive RN excess movement, this is not necessarily the case for all possible DGPs. Taking the envelope over all of these processes for $\phi = 3$ yields the bound from Proposition 2, which is shown in the thick blue line. It is asymmetric around 0.5 as well as nonmonotonic in π_0^*.[12] Finally, the thick red line shows the bound for the limiting case $\phi \to \infty$, which is equal to the squared RN prior from Corollary 1.

[11] In the limit as $\phi \to \infty$, $\pi_0 \to 0$ for any π_0^*, so $\pi_0^* - \pi_0 \to \pi_0^*$.

[12] The second term in the bound (π_0^*) of course increases monotonically in π_0^*, generating asymmetry in the bound around $\pi_0^* = 0.5$ for $\phi > 1$. But the first term in the bound $(\pi_0^* - \pi_0)$ does not increase monotonically for $1 < \phi < \infty$, exerting a countervailing force that generates the observed nonmonotonicity.

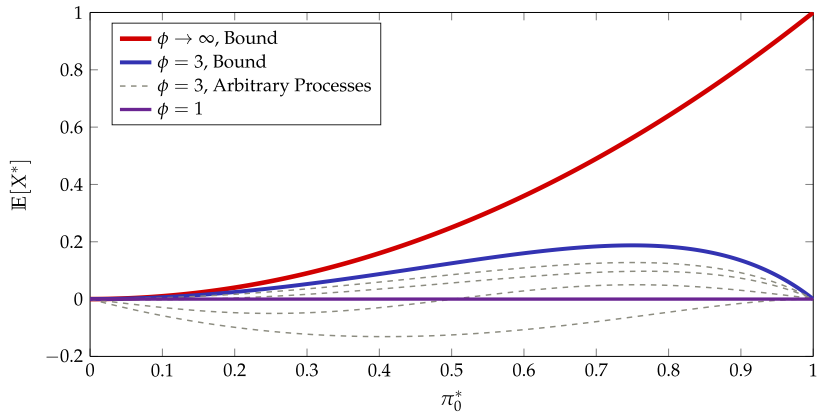


Figure 3. RN excess belief movement versus prior by ϕ under RE. Theoretical bounds are obtained from the formulas in Proposition 2 and Corollary 1. (Color figure can be viewed at wileyonlinelibrary.com)

C.2. Additional Results

The Internet Appendix contains a number of additional theoretical results (Section II.A of the Internet Appendix) and simulations (Section II.B of the Internet Appendix), which we summarize briefly here. First, we characterize how the DGP determines the sign and magnitude of RN excess movement $\mathbb{E}[X^*]$. Given that we observe positive excess movement in the data, most of our focus in the main text is on the positive upper bounds for $\mathbb{E}[X^*]$. But as shown in Propositions IA1 and IA2, asset-pricing settings with risk aversion do not necessarily entail positive excess movement in observed beliefs. Further, given the arbitrary labeling of the two states, there is no a priori reason to expect under RE that $\mathbb{E}[X^*]$ should take a particular sign.

Second, we conduct numerical simulations to further explore how RN excess movement depends on the DGP. When the DGP for beliefs is symmetric (with equal movements up or down, as is likely to roughly apply for options), we find that $\mathbb{E}[X^*]$ takes the same sign as $\pi_0^* - 0.5$. This suggests, for example, that for priors close to 0.5, excess movement should be small unless the DGP is asymmetric. Indeed, we also find that extreme values of $\mathbb{E}[X^*]$ occur only in highly asymmetric DGPs where movements in one direction are large and movements in the other direction are tiny.

Finally, in Proposition IA3, we show that the bound in Proposition 2 is approximately tight: There exists a DGP for which $\mathbb{E}[X^*]$ is close to the bound for large T. The proof is constructive, as we characterize the form of the maximizing DGP. In line with the simulations, this DGP takes a somewhat perverse form, indicating further that the bound is quite conservative.

II. Generalized Theoretical Results for Equilibrium Asset Prices

While the binary-state setting considered in the previous section is useful for exposition, it is also artificial: One cannot obtain a single person's valuation of

Arrow-Debreu claims from observational data alone; there are more than two possible states; and the realized state determines more than just consumption. We thus consider a general many-state framework for equilibrium asset prices and show how our results extend to this empirically relevant case.

A. Setup and Notation

A.1. Preliminaries: Probability Space, Prices, and Risk-Neutral Probabilities

Time is again indexed by $t \in \{0, 1, 2, \ldots\}$, and we consider a discrete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $\{H_t\}$.[13] A realization of the elementary state is denoted by $\omega \in \Omega$. To make our results empirically implementable, we will focus on payoffs tied to the spot price of the market index, $S_t : \Omega \to \mathbb{R}_+$, on some option expiration date T. (We extend the notation below to allow for multiple option expiration dates.) A European call option on the index with strike price K has date-T payoff $(S_T - K)^+ = \max(S_T - K, 0)$, and its time-$t$ price is $q_{t,K}$. These option prices are observable for some set of strike prices $\mathcal{K} \subseteq \mathbb{R}_+$ beginning at date 0. Assuming the absence of arbitrage, there exists a strictly positive SDF $M_{t,T}$ such that option prices satisfy $q_{t,K} = \mathbb{E}_t[M_{t,T}(S_T - K)^+]$, where $\mathbb{E}_t[\cdot] \equiv \mathbb{E}^{\mathbb{P}}[\cdot \mid H_t]$. The SDF can equivalently be written as $M_{t,T} = M_T/M_t$ for strictly positive $\{M_t\}$, with $M_0 = 1$.

Option prices will be of interest for inferring a distribution over the change in value of the market index from 0 to T (rather than consumption, for which options are not directly traded). We say that the *return state* $\theta \in \Theta \subset \mathbb{R}_+$ is realized for the market as of date T if $R_T \equiv S_T/S_0 = \theta$. The measure $\mathbb{P} : \mathcal{F} \to [0, 1]$ governs the *objective* or *physical* probabilities of these return states.[14] In this general case, the *risk-neutral (RN) measure* is defined by the change of measure

$$\left.\frac{d\mathbb{P}^*}{d\mathbb{P}}\right|_{H_t} = \frac{M_{t,T}}{\mathbb{E}_t[M_{t,T}]} = \frac{M_T}{\mathbb{E}_t[M_T]}, \tag{11}$$

and expectations under $\mathbb{P}^*$ are denoted by $\mathbb{E}^*[\cdot]$.[15] Using this definition of $\mathbb{P}^*$, the RN probability of return state θ is

$$\mathbb{P}^*_t(R_T = \theta) = \frac{\mathbb{E}_t[M_T \mid R_T = \theta]}{\mathbb{E}_t[M_T]}\,\mathbb{P}_t(R_T = \theta). \tag{12}$$

The RN pricing equation $q_{t,K} = \mathbb{E}^*_t[(S_T - K)^+]/R^f_{t,T}$ can be used to show that the date-t option prices $\{q_{t,K}\}_{K \in \mathcal{K}}$ reveal the set of RN probabilities

[13] We could consider continuous states with additional technicalities.

[14] It is without loss of generality to define this measure over arbitrary fixed return states: We are simply discretizing the set of possible index values as of expiration date T and keeping track of the distribution over these fixed outcomes.

[15] As $\text{Price}_t = \mathbb{E}_t[M_{t,T}\text{Payoff}_T]$ for any asset, we have $R^f_{t,T} = \mathbb{E}_t[M_{t,T}]^{-1}$, where $R^f_{t,T}$ is the gross risk-free rate from t to T, and $\text{Price}_t = \mathbb{E}^*_t[\text{Payoff}_T]/R^f_{t,T}$. This $\mathbb{P}^*$ is sometimes referred to as the T-forward measure.

$\{\mathbb{P}_t^*(R_T = \theta)\}_{\theta\in\Theta}$. Assume that the set of return states Θ is ordered such that $\theta_1 < \theta_2 < \cdots < \theta_J$, and assume for notational simplicity that the set of traded option strikes $\mathcal{K}$ coincides with the set of possible date-T index values (i.e., $\mathcal{K} = \{K_j\}_{j=1}^J$, with $K_j = S_0\theta_j$). We can then back out RN probabilities from option prices as follows:

$$\mathbb{P}_t^*(R_T = \theta_j) = R_{t,T}^f\left[\frac{q_{t,K_{j+1}} - q_{t,K_j}}{K_{j+1} - K_j} - \frac{q_{t,K_j} - q_{t,K_{j-1}}}{K_j - K_{j-1}}\right]. \tag{13}$$

See Internet Appendix Section I.A for a derivation of this result, which follows from a discrete-state application of the classic result of Breeden and Litzenberger (1978). As in Section I.B, the above RN probabilities are a more convenient object of analysis than raw option prices.

A.2. Beliefs

Aside from assuming no arbitrage, we have not yet taken a stance on the market structure or subjective beliefs underlying prices and RN probabilities. We could in principle pursue a strict mapping from Section I to the current case, by assuming a setting in which all individual traders have common beliefs satisfying RE. But our framework accommodates less strict requirements as well. As long as there is an unconstrained agent with rational beliefs satisfying our three main assumptions below, then the tests we derive should not be rejected in the data. One can therefore think of our framework as testing for the possible existence of such an agent.[16] More loosely, one might also consider an interpretation in which the "agent" in question is the market as a whole.

Regardless of the interpretation of the agent in question, assume that prices correspond to valuations for an agent who, at the beginning of each period, observes a signal vector $s_t \in \mathcal{S}$ drawn from the distribution $DGP(s_t \mid \theta, H_{t-1}) = \mathbb{P}_{t-1}(s_t|\theta)$, where θ is the return state realized at T and $H_t = \sigma(s_\tau, 0 \leqslant \tau \leqslant t)$ is the σ-algebra generated by the history of signal realizations. The agent's subjective belief distribution over return states is denoted by $\Pi_{t,T} = \{\pi_t(R_T = \theta)\}_{\theta\in\Theta}$, where $\pi_t(R_T = \theta) \geqslant 0\ \forall\, \theta \in \Theta$ and $\sum_{\theta\in\Theta} \pi_t(R_T = \theta) = 1$. More generally, for any random variable $Y(\omega)$, the agent attaches subjective probability $\pi_t(Y = y)$ to the outcome $Y = y$. We generalize Assumption 1 as follows.

ASSUMPTION 2 (RE): *For any random variable Y, beliefs satisfy $\pi_t(Y = y) = \mathbb{P}_t(Y = y)$ with probability one for all t.*

This assumption again implies that beliefs satisfy $\pi_t(R_T = \theta) = \mathbb{E}[\pi_{t+1}(R_T = \theta) \mid \pi_t(R_T = \theta)]$ for all $\theta \in \Theta$. As in Section I, this martingale condition for beliefs over returns is all that is required for our main results to carry through. The full-RE generalization stated in Assumption 2 is useful for streamlining some of the remaining discussion, as it further implies that *all* conditional expectations—including over the SDF—are martingales with respect to H_t.

[16] We thank an anonymous referee for suggesting this point.

Given Assumption 2, we can define the RN belief distribution without explicitly restricting the agent's utility or constraint set by applying the same change of measure as in (11). This yields the RN belief distribution $\Pi^*_{t,T} = \{\pi^*_t(R_T = \theta)\}_{\theta\in\Theta}$ such that $\pi^*_t(R_T = \theta) = \frac{\mathbb{E}_t[M_T \mid R_T=\theta]}{\mathbb{E}_t[M_T]}\pi_t(R_T = \theta)$ as in (12), and thus (13) tells us that option prices reveal the agent's RN beliefs as given here.

A.3. Localization: Conditional Beliefs, SDF Ratio, and Excess Movement

To align with the analysis in Section I, we consider the behavior of *conditional* RN beliefs over adjacent pairs of return states. That is, rather than directly considering the full distribution $\Pi^*_{t,T}$, we instead consider restrictions on the behavior of the binarized beliefs $\{\widetilde{\pi}^*_{t,j}\}_{j=1}^{J-1}$, defined by

$$\widetilde{\pi}^*_{t,j} \equiv \pi^*_t(R_T = \theta_j \mid R_T \in \{\theta_j, \theta_{j+1}\}) = \frac{\pi^*_t(R_T = \theta_j)}{\pi^*_t(R_T = \theta_j) + \pi^*_t(R_T = \theta_{j+1})},$$

for $\pi^*_t(R_T = \theta_j) + \pi^*_t(R_T = \theta_{j+1}) > 0$. In words, $\widetilde{\pi}^*_{t,j}$ is the RN belief that return state θ_j will be realized, conditional on either θ_j or θ_{j+1}. This binary localization will be useful for two reasons: (i) It will allow us to apply results from the two-state setting of Section I, and (ii) the main identifying assumption used to derive our tests is less restrictive than it would be without such a transformation (as discussed in Section III.A below). Conditional physical beliefs $\widetilde{\pi}_{t,j}$ are defined analogously, and the expectation under the conditional physical measure $\widetilde{\mathbb{P}}_t$ is $\widetilde{\mathbb{E}}_t[\,\cdot\,] \equiv \mathbb{E}_t[\,\cdot \mid R_T \in \{\theta_j, \theta_{j+1}\}]$.

In this context, the analogue to ϕ as defined in Section I is

$$\phi_{t,j} \equiv \frac{\mathbb{E}_t[M_T \mid R_T = \theta_j]}{\mathbb{E}_t[M_T \mid R_T = \theta_{j+1}]}, \tag{14}$$

which encodes the slope of the SDF across the adjacent return states θ_j, θ_{j+1}. In a representative-agent economy with SDF $M_{t,T} = \beta^{T-t}\frac{U'(C_T)}{U'(C_t)}$, the above definition implies $\phi_{t,j} = \frac{\mathbb{E}_t[U'(C_T) \mid R_T=\theta_j]}{\mathbb{E}_t[U'(C_T) \mid R_T=\theta_{j+1}]}$, akin to (6). But (14) is general and does not require a representative-agent structure (although we make periodic reference to such an economy for interpretation). This SDF slope can also be thought of loosely as the local price of risk around index return state θ_j.

Using this definition of $\phi_{t,j}$, RN beliefs satisfy $\frac{\widetilde{\pi}^*_{t,j}}{1-\widetilde{\pi}^*_{t,j}} = \phi_{t,j}\frac{\widetilde{\pi}_{t,j}}{1-\widetilde{\pi}_{t,j}}$ and thus $\widetilde{\pi}^*_{t,j} = \frac{\phi_{t,j}\widetilde{\pi}_{t,j}}{1+(\phi_{t,j}-1)\widetilde{\pi}_{t,j}}$, as in (5). Given a resolving RN belief stream $\boldsymbol{\pi}^*_j = [\widetilde{\pi}^*_{0,j}, \ldots, \widetilde{\pi}^*_{T,j}]$, we define RN belief movement m^*_j, RN initial uncertainty $\mathsf{u}^*_{0,j}$, and RN excess movement X^*_j as in (1) to (3), with $\widetilde{\pi}^*_{t,j}$ in place of π_t. We often suppress the dependence on j (writing X^*, for example) when considering an arbitrary state pair.

B. Identifying Assumptions on ϕ

The analysis in Section I considered Arrow-Debreu claims on primitive states (in that case, consumption states). This section's analysis instead considers claims on index-return states, with an eye toward empirical implementation. We must therefore confront the joint hypothesis problem. The additional identifying assumptions to be tested jointly with Assumption 2 take the form of two restrictions on $\phi_{t,j}$. We introduce these assumptions here, and we discuss them in much greater detail in the following sections after presenting our main results below.

We first impose an ordering assumption on the states. Return state θ_j here corresponds to state 1 in Section I (versus state 0 for θ_{j+1}). We thus maintain the convention of labeling θ_j as the "bad" state, so that $\phi_{t,j} \geqslant 1$. While this is an innocuous labeling convention in theory, empirical implementation requires taking a stand on how to distinguish θ_j from θ_{j+1}. We make the intuitive assumption that the bad state θ_j corresponds to the lower return.

ASSUMPTION 3 (POSITIVE RISK AVERSION IN INDEX RETURN): *$\phi_{t,j} \geqslant 1$ with probability one for all t, j, where the set of return states Θ is ordered such that $\theta_1 < \theta_2 < \cdots < \theta_J$.*

We discuss the empirical plausibility of this assumption, and how our results can be modified if it is violated (as might be a concern given the so-called pricing kernel puzzle), in Section III.D.

Second, and more substantively, we must impose an assumption on the evolution of the unobserved parameter $\phi_{t,j}$. In Section I's setting, ϕ is naturally constant over t: The terminal states index consumption and thus marginal utility, so the marginal rate of substitution across these primitive states is fixed. For our main analysis, we impose an analogous assumption on $\phi_{t,j}$.

ASSUMPTION 4 (CONSTANT SDF RATIO, OR CONDITIONAL TRANSITION INDEPENDENCE): *We say that the SDF satisfies* conditional transition independence (CTI) *for return-state pair (θ_j, θ_{j+1}) if $\phi_{t,j}$ is constant with probability one for all t. We assume that CTI is satisfied for all* interior *state pairs, $j = 2, 3, \ldots, J-2$, and we write $\phi_{t,j} = \phi_j$ for these states.*

This assumption imposes that the *relative* expected "severity" of the adjacent return states is constant over a contract, so that the expected SDF (or marginal utility) realization in the low-return state θ_j is a constant multiple of the expected realization in state θ_{j+1}. This is akin to assuming a constant local price of risk, as discussed further in Section III. It corresponds to a notion of transition independence because it implies that the expected relative SDF realizations depend only on the return-state pair and not on the path of variables realized between t and T (see Lemma A3 in the Appendix). Note that we exclude the extreme state pairs (θ_1, θ_2) and (θ_{J-1}, θ_J) from the constant-ϕ requirement: Thinking of θ_1 and θ_J as tail returns, we are allowing for time-varying disaster (or positive jump) risk.

As we discuss in Section III, this assumption is weaker than the joint assumptions imposed in past volatility tests, and we view it as a reasonable starting point for our analysis. It is unlikely to hold perfectly in reality, however, so Section III also provides results characterizing RN excess movement when the assumption is violated.

C. Main Results in the General Setting

Having completed the formal setup and description of our main assumptions, we turn now to our main results in this more general asset-pricing setting. The bulk of the work in this case is in the setup and notation, as all of our main results apply with appropriate relabeling.

PROPOSITION 3: *Under no arbitrage and Assumptions 2 to 4, for* $j = 2, 3, \ldots, J-2$*, the results stated in Lemma 1, Propositions 1 and 2, and Corollary 1 continue to hold, with* $\widetilde{\pi}^*_{t,j}$ *replacing* π^*_t*,* $\widetilde{\pi}_{t,j}$ *replacing* π_t*,* X^*_j *replacing* X^**,* ϕ_j *replacing* ϕ*,* $\widetilde{\mathbb{E}}_0[\cdot]$ *replacing* $\mathbb{E}[\cdot]$*, and* $\triangle_j \equiv \widetilde{\mathbb{E}}_0[X^*_j \mid R_T = \theta_{j+1}] - \widetilde{\mathbb{E}}_0[X^*_j \mid R_T = \theta_j]$ *replacing* $\triangle$*.*

The main theoretical complication in applying the results in Section I to this setting is in proving that $\widetilde{\mathbb{E}}^*[X^*_j \mid R_T = \theta_j] = \mathbb{E}[X^*_j \mid R_T = \theta_j]$. The economic intuition for these results is largely identical to that discussed in Section I. And while the results above are convenient to express in terms of the SDF slope ϕ_j given that this allows for closed-form solutions that can be applied regardless of the origin of the SDF, the results also admit an interpretation in terms of the approximate required risk aversion for a fictitious representative agent who consumes the market index (analogous to (7)). To avoid repetition, for the remaining results we will continue to assume no arbitrage and that Assumptions 2 to 4 hold for $j = 2, 3, \ldots, J-2$, unless stated otherwise.

PROPOSITION 4: *Assume additionally that there is a representative agent with (indirect) utility over time-T wealth, with wealth equal to the market index value, and denote* $S_j \equiv S_0\theta_j$*. Then local relative risk aversion* $\gamma_j \equiv -S_jU''(S_j)/U'(S_j)$ *is given to first order around return state* θ_j *by*

$$\gamma_j = \frac{\phi_j - 1}{(S_{j+1} - S_j)/S_j}.$$

This result formalizes the sense in which ϕ_j reflects the local price of risk, as it corresponds to the market's effective relative risk aversion in a neighborhood around return θ_j. As in Section I, γ_j is proportional to $\phi_j - 1$, as this gives the percent decrease in marginal utility in moving from low-return state θ_j to high-return state θ_{j+1}. To calculate relative risk aversion, this change in marginal utility must be normalized by the percent wealth increase $(S_{j+1} - S_j)/S_j$ in moving from θ_j to θ_{j+1}, which is also equal to the percent return deviation $(\theta_{j+1} - \theta_j)/\theta_j$ between the two states. If, for example, $\theta_j = 1$, $\theta_{j+1} = 1.05$, then $\phi_j = 1.5$ implies $\gamma_j = 10$.

III. Interpreting and Relaxing the Test's Assumptions

Our analysis thus far has proceeded under the joint null implied by Assumptions 2 to 4, which we now discuss in more detail. What do these assumptions entail specifically? How informative is a rejection of the joint null, and how should it be interpreted? We begin by considering CTI, which we consider the most important assumption imposed alongside RE. We first discuss settings in which it does (and does not) hold, provide a theoretical result relaxing the assumption, and conduct simulations of models in which it does not hold. We then briefly discuss the two other assumptions and provide a result accounting for possible mismeasurement or microstructure noise.

A. CTI Generates an Informative Test

We note first that the constant-ϕ assumption is significantly weaker than the assumption of constant discount rates. Our framework allows for any variation in the physical distribution of R_T, which can be thought of as changes in the *quantity* of risk. Further, by considering RN beliefs implied by options, we allow arbitrary variation in risk-free discount rates (since risk-free discounting affects all option prices proportionally). Both sources of variation are ruled out in tests with constant discount rates. We formalize these statements and discuss the relationship between RN beliefs and discount rates further in Internet Appendix Section II.C.

That said, the assumption of constant ϕ is substantively restrictive in one way: the price of risk must be locally constant over the life of a given option contract. We are assuming, for example, that the market's effective risk aversion over whether the return will be 0% or 5% is fixed over a span of weeks or months. Thus, a violation of our bounds can be interpreted as reflecting either excess volatility in the marginal investor's beliefs or variation in this local price of risk. A constant local price of risk, however, does not require that the *aggregate* price of risk is constant. For example, if risk aversion is decreasing in R_T, then bad news about the return distribution can increase aggregate risk aversion without affecting *local* risk aversion in a neighborhood of a 0% return.

More generally, CTI is satisfied in a range of theoretical settings, with a few notable exceptions. We make this point not to assert that CTI must hold in reality, but instead to demonstrate that a violation of our bounds is informative in ruling out a range of well-studied frameworks:[17]

(i) Any shocks that affect the expectation of M_T proportionally in all return states (e.g., multiplying $\mathbb{E}_t[M_T \mid R_T = \theta_j]$ by 1.2 for all j) are trivially ruled in by the definition of CTI. More generally, if $M_T = \delta_T f(R_T)$ for some function $f(\cdot)$, and shocks to δ_T are orthogonal to $f(R_T)$, then CTI holds. This rules in at least a subset of permanent SDF shocks as

[17] Derivations for statements (iii) and (iv) and the subsequent examples are in Internet Appendix Section I.B. Statements (i) and (ii) are immediate.

classified by Alvarez and Jermann (2005), although we note that it does not cover all possible permanent shocks.[18]

(ii) If a representative agent's utility depends only on the terminal index value, then CTI holds. Similarly, if there exists an agent whose indirect utility can be written as a function only of the time-T index value—for example, an unconstrained investor with horizon T (i.e., Utility $=$ $f(\text{Wealth}_T)$) who is fully invested in the market—then CTI holds. This includes settings in which an unconstrained log investor holds the market, which is the leading case studied by Martin (2017) in measuring the equity premium (see also Gandhi, Gormsen, and Lazarus, 2026).

(iii) In the variable rare-disasters model of Gabaix (2012), CTI holds for all market return-state pairs (θ_j, θ_{j+1}) for which there is negligible probability of having realized a disaster conditional on reaching θ_j.[19] This illustrates the usefulness of the localization provided by considering conditional beliefs: If we are concerned that time-varying disaster risk may affect $\phi_{t,j}$ for states in the tail of the return distribution, we can ignore these states and confine attention to the center of the return distribution, where $\phi_{t,j}$ can be expected to be approximately constant.

(iv) If a representative agent has Epstein and Zin (1989) utility and holds the market, then CTI holds if (i) relative risk aversion is $\gamma = 1$, and the elasticity of intertemporal substitution (EIS) ψ and DGP are arbitrary; (ii) γ is arbitrary, the EIS is $\psi = 1$, and log consumption growth follows an AR(1) with normal innovations; or (iii) γ and ψ are arbitrary, and consumption growth is i.i.d. (a less interesting case). As Martin (2017, p. 378) notes, case (i) is "considered (and not rejected) by Epstein and Zin (1991) and Hansen and Jagannathan (1991)." Case (ii) approximates Case I of the Bansal and Yaron (2004) long-run risks model (without stochastic volatility), with $\psi = 1$ and common shocks to current and expected future consumption. Finally, case (iii) is considered by Martin (2013).

These examples illustrate further that the assumption of CTI is meaningfully weaker than the assumption of constant discount rates. This should again not be taken as evidence *in favor* of CTI holding; instead, it allows for an informative joint test whose null includes a range of models that have been advanced as rationalizations of the excess-volatility puzzle.

That said, there are also well-known models under which CTI does *not* hold. As suggested by statement (iv), models featuring priced stochastic volatility generally pose an issue for CTI. Since priced volatility risk is a central feature

[18] Meanwhile, Borovička, Hansen, and Scheinkman (2016) show that permanent SDF shocks are ruled out in the framework proposed by Ross (2015) for recovering physical probabilities from state prices.

[19] Formally, for any $\delta>0$, $\exists\, \underline{\theta}$ s.t. $\forall\, \theta_j \geqslant \underline{\theta}$, $\mathbb{P}_0\left(\sum_{t=1}^{T} \mathbb{1}\{\text{disaster}_t\} > 0 \,\middle|\, R_T \geqslant \underline{\theta}\right) < \delta$, so the conditional probability of having realized a disaster is negligible. For all $\theta_j \geqslant \underline{\theta}$, CTI holds for the state pair (θ_j, θ_{j+1}) up to a negligible error.

of most leading option pricing models, we conduct a detailed simulation study of how much RN belief variation such models generate in Section V.A.

Other models in which CTI is violated further illustrate its content. CTI fails in the habit model of Campbell and Cochrane (1999): The path of consumption matters in a manner not fully accounted for by conditioning on the return state. In addition, CTI can fail to hold in heterogeneous-belief models if there is no rational investor who is unconstrained with probability one. For example, in the Basak (2000) model with heterogeneous beliefs and extraneous (nonfundamental) risk, CTI fails to hold if extraneous risk is perceived to matter by all agents. We view the fact that CTI rules out stochastic volatility and habit models to be a downside of the framework, which we address with robustness results below. The fact that we rule out some (though not all) models of belief volatility induced by heterogeneity, however, is a feature: Time variation in ϕ in these cases is equivalent to the as-if representative agent having excessively volatile beliefs, which represents a meaningful alternative to our null.

B. Relaxing CTI: Theory and Simulations

While CTI allows for an informative null in our main analysis, it is also a knife-edge restriction that is unlikely to hold exactly. We therefore now investigate how such a violation affects RN excess movement. One might be concerned that even small fluctuations in ϕ could generate dramatic violations of our bounds. For example, suppose that π_t is constant at 0.5 and ϕ_t changes back and forth repeatedly from 1 to 1.5. Without any movement in physical beliefs, π_t^* will vary repeatedly between 0.5 and 0.6, leading to unbounded movement as $T \to \infty$. But this argument overlooks a core insight: this repeated oscillation is *also* inconsistent with RE as long as the variance of ϕ_t is bounded, because it implies predictable mean reversion in the expected marginal value of \$1. That is, the fact that ϕ_t is itself a function of martingale conditional expectations restricts its evolution under RE. We pursue this logic more formally here, both theoretically and in a set of simulations.

First, theoretically, dropping the constant-ϕ_t assumption comes at the cost of much of the parsimony in our analysis: SDF expectations nonlinearly map to ϕ_t, which nonlinearly combines with π_t to determine π_t^*, which in turn is nonlinearly mapped to X^*. Perhaps surprisingly, however, we can still make theoretical statements for a meaningful subset of DGPs. Consider ϕ_t as defined in (14) (continuing to suppress j subscripts). The agent now also receives signals $s_{\phi_t} \in \mathcal{S}_{\phi_t}$ to learn about ϕ_t over time. For simplicity, assume that uncertainty over ϕ_t evolves on a binomial tree ($|\mathcal{S}_{\phi_t}| = 2$), although this is not necessary for the following statements. We also assume that π_t and ϕ_t do not change (relative to their $t-1$ values) together in the same period.[20] The following result then holds.

[20] In a previous version of this paper (Augenblick and Lazarus, 2022), we prove the result for arbitrary $|\mathcal{S}_{\phi_t}|$. For the second assumption, one can split each period into two subperiods, with ϕ_t changing in the first half and π_t in the second. The proof becomes intractable without this assumption, but we provide simulation evidence relaxing it below.

PROPOSITION 5: *If ϕ_t evolves as a martingale or supermartingale ($\mathbb{E}_t[\phi_{t+1}] \leqslant \phi_t$) and $Var(\phi_t) < \infty$, then the bounds in Proposition 2, Corollary 1, and Proposition 3 continue to apply, with ϕ_0 replacing ϕ.*

The proof of Proposition 5 demonstrates that the variation in π_0^* arising when ϕ_t is a nondegenerate (super)martingale always strictly lowers excess RN belief movement, rendering the main bound in (10) even more conservative. Changes in ϕ_t do cause π_t^* to change, which adds movement. But when ϕ_t is a supermartingale, this movement works against the upward bias in RN beliefs themselves and decreases $\pi_t^* - \pi_t$ in expectation. This reduces potential future excess movement more than enough to offset the increase in period t.

As for the interpretation of the supermartingale restriction, from the definition of covariance, $\phi_{t,j}$ is a supermartingale if and only if $\text{Cov}_t(\phi_{t+1,j}, \mathbb{E}_{t+1}[M_T \mid R_T = \theta_{j+1}]) \geqslant 0$. Interpreting the SDF M_T as proportional to $U'(C_T)$, this requires that risk aversion (encoded in $\phi_{t+1,j}$) be positively correlated with expected marginal utility (MU) in the high-C_T state. This is an intuitively reasonable restriction, as it implies that bad news (higher expected MU) generally arrives at the same time for both states; see Lazarus (2022) for further discussion. That said, the requirement that π_t and ϕ_t do not vary contemporaneously may be more restrictive. And given the complexity of the setting, it is difficult to make analytical statements when ϕ_t does not satisfy the assumptions of Proposition 5.

We therefore next turn to a set of simulations in which ϕ_t can vary with more freedom. For now, we consider a set of abstract DGPs. We also conduct simulations of specific option pricing models with varying ϕ_t; we compare these quantitatively with our empirical results, and accordingly postpone discussion of these simulations to the next section. In our abstract simulations, the agent learns over time about the terminal state and the SDF realization in both the j and $j+1$ states. Each of these three objects takes one of two values at T, with different combinations of ex ante uncertainty and signal strengths for all three objects. We outline the main results of these simulations here, and we describe the setting further and present full results in Internet Appendix Section II.D.

The simulations all start with $\pi_0^* = 0.5$ and $\phi_0 = 3$ but differ in the variation in ϕ_t, with the ex ante standard deviation of ϕ_T ranging from $\text{SD}_0(\phi_T) = 0$ to 3. Using Proposition 4, if states θ_j and θ_{j+1} differ by 5% (as in our data), this corresponds to relative risk aversion of $\gamma_0 = 40$ and a standard deviation for γ_T between 0 and 60. Figure IA.2 plots distributions for the estimated $\mathbb{E}[\text{m}^*]$ across simulations. It shows that uncertainty in ϕ_T has limited impact on estimated movement in these DGPs. Even in the case with the highest uncertainty in ϕ_T, movement $\mathbb{E}[\text{m}^*]$ increases only by 0.015, or by 6% relative to the constant-ϕ value of $\mathbb{E}[\text{m}^*] = \text{u}_0^* = 0.25$. Further, $\mathbb{E}[X^*]$ rarely exceeds the bound for $\phi = 3$ (in just a fraction of 1% of the high-uncertainty simulations). And in all cases, the bound in Corollary 1 for $\phi \to \infty$ holds for 100% of the simulations.

We conclude, somewhat surprisingly, that even significant uncertainty in ϕ_T does not meaningfully affect our bounds, at least for these DGPs. When the agent updates her SDF expectations, these updates must still respect RE. As

a result, even large values for σ_ϕ do not allow for arbitrary oscillations in ϕ_t, as information about ϕ_T is revealed gradually over time. That said, these binary DGPs are somewhat difficult to map to more fully specified option pricing models, which may produce richer time variation in ϕ_t than considered in these preliminary simulations. We therefore return to simulations of these option pricing models after presenting our empirical results.

C. Aggregating over Belief Streams

The previous subsection considers the effect of time variation in ϕ_j *within* a given RN belief stream (i.e., within a single option contract, lasting weeks or months). But even if the effects of such higher-frequency variation are small, it may be less palatable to assume that ϕ_j is constant *across* belief streams: There may be lower-frequency variation over the span of years. We now ask how such variation affects our bounds when considering multiple belief streams.

Answering this question is also important for empirical implementation. The bounds in Proposition 3 are stated as date-0 expectations conditional on the RN prior, but we observe only one draw X_j^* per expiration date rather than the expectation of this statistic for a given $\widetilde{\pi}_{0,j}^*$. Estimation thus requires aggregating over multiple streams with different $\widetilde{\pi}_{0,j}^*$ and potentially different ϕ_j, especially since our estimation groups different streams together using the same fixed return states.

Thus, generalizing slightly, assume now that we can observe index options for N expiration dates $\mathcal{T} \equiv \{T_i\}_{i=1}^N$, so i indexes belief streams (and their DGPs). We use $\phi_{i,j}$ for the SDF ratio for expiration date T_i and state pair (θ_j, θ_{j+1}); RN beliefs are $\widetilde{\pi}_{t,i,j}^*$; and RN excess movement is $X_{i,j}^*$. We again often suppress j for an arbitrary state pair. Due to Jensen's inequality, we cannot simply use $\mathbb{E}[\phi_i]$ in place of ϕ_i or $\mathbb{E}[\pi_{0,i}^*]$ in place of $\pi_{0,i}^*$ when taking the expectation of both sides of the results in Propositions 1 to 3 over all i. However, the following generalizations do hold.

PROPOSITION 6: *Define* $\overline{\phi} \equiv \max_{\pi_{0,i}^*} \mathbb{E}[\phi_i \mid \pi_{0,i}^*]$. *We have:*

(i) GENERALIZATION OF PROPOSITION 1*: If* $Cov(\pi_{0,i}, \triangle_i) = 0$*, and* $\pi_{0,i}^*$ *is constant across* i *(i.e., fixing a given* $\pi_{0,i}^*$*), then over all streams,*

$$\mathbb{E}[X_i^*] \leqslant \max\left\{0, \left(\pi_{0,i}^* - \frac{\pi_{0,i}^*}{\mathbb{E}[\phi_i] + (1 - \mathbb{E}[\phi_i])\pi_{0,i}^*}\right)\mathbb{E}[\triangle_i]\right\}. \tag{15}$$

(ii) GENERALIZATION OF PROPOSITION 2*: Over all streams, without any additional assumptions,*

$$\mathbb{E}[X_i^*] \leqslant \mathbb{E}\left[\left(\pi_{0,i}^* - \frac{\pi_{0,i}^*}{\overline{\phi} + (1 - \overline{\phi})\pi_{0,i}^*}\right)\pi_{0,i}^*\right], \tag{16}$$

or, fixing a given $\pi^*_{0,i}$, $\mathbb{E}[X^*_i] \leqslant \left(\pi^*_{0,i} - \frac{\pi^*_{0,i}}{\mathbb{E}[\phi_i]+(1-\mathbb{E}[\phi_i])\pi^*_{0,i}}\right)\pi^*_{0,i}$.

(iii) GENERALIZATION OF COROLLARY 1*: Over all streams, without any additional assumptions,*

$$\mathbb{E}[X^*_i] \leqslant \mathbb{E}\left[{\pi^*_{0,i}}^2\right].$$

Only the analogue to Proposition 1 requires an additional assumption. The original bound is $(\pi^*_0 - \pi_0)\triangle$, so the covariance between $\pi_{0,i}$ and $\triangle_i$ over DGPs affects the generalized bound. For simplicity, we set this covariance to zero.[21] Part (ii), meanwhile, generalizes (10); as can be seen in the proof of the result, it is more conservative than the original bound for a single stream. Finally, the bound that does not depend on $\phi_{i,j}$ (part (iii)) applies as previously stated.

These bounds are now empirically implementable, and the minimum $\overline{\phi}$ that solves (16) is a conservative estimate of the maximal conditional-mean SDF slope for the return-state pair in question. Finally, reintroducing dependence on the state pair j, it is likely that the values $\overline{\phi}_j$ vary over j. But the same steps used for Proposition 6 to take expectations over i can also be applied to take expectations over j, thereby yielding a single estimate $\overline{\boldsymbol{\phi}}$ aggregated over both streams *and* return states (for all states meeting CTI) when desired.

D. Robustness to Additional Assumptions

Having considered Assumption 4 in detail, we now turn to the two other assumptions to continue our discussion of how to interpret a rejection of the joint null. To conserve space, we relegate details to Internet Appendix Section II.E and provide a brief summary here.

First, considering Assumption 2, this assumption requires both a correct subjective prior and rational updating using the true signal distribution as the likelihood. A natural question is whether an incorrect prior by itself can generate violations of the upper bound for X^*, or whether excessive movement requires incorrect updating. We show in Proposition IA4 that because an incorrect prior acts as a one-time belief distortion, it is generally not sufficient for a full violation of the bound in Proposition 2. In general, then, incorrect updating behavior must be present in such a violation.[22]

We now turn to Assumption 3, which imposes $\phi_j \geqslant 1$. This requires that the expected SDF realization in the low-return state be greater than in the high-return state. A line of work beginning with Jackwerth (2000), however, argues that the SDF does not decrease monotonically with the index return in options

[21] This is equivalent to assuming no relationship between the asymmetry of the DGP and ϕ. This part also holds fixing $\pi^*_{0,i}$; this is sufficient for us, as our empirical results for this bound will generally be conditional on a given $\pi^*_{0,i}$.

[22] Note that even if ϕ is constant under the agent's subjective belief, incorrect updating will induce a probability distortion such that the *actual* SDF (assessed under $\mathbb{P}$) will feature a time-varying ϕ_t.

data. This finding, often referred to as the "pricing kernel puzzle," would imply a violation of Assumption 3 for at least some subset of the return space. Given such a violation, it turns out that there is still a well-defined theoretical upper bound for RN excess movement: We show in Proposition IA5 that our main bounds apply with minor modification (effectively flipping the role of the two states when $\phi < 1$), and for *any* value of ϕ, $\mathbb{E}[X^*] \leqslant \max(\pi_0^{*2}, (1-\pi_0^*)^2)$. So excess movement is bounded regardless of ϕ. While this is useful theoretically, the bound in the $\phi < 1$ case would have to be estimated separately from our main bounds derived for the $\phi \geqslant 1$ case. So for the sake of simplicity, we empirically estimate only the main bounds. We do so in part because we estimate the bounds separately for multiple points in the return space. Thus, even if one is concerned about the pricing kernel puzzle affecting our results, one can confine empirical attention to our estimates for return ranges for which the puzzle does not emerge.[23]

E. Robustness to Measurement Error

The robustness results above speak to the possibility of theoretical misspecification. As the final step in making our bounds implementable, we now consider how to account for possible *empirical* misspecification arising from mismeasurement or microstructure noise in RN beliefs. The bounds provide a minimum value of ϕ required to rationalize the observed variation in RN beliefs; if some of this variation is in fact arising due to noise, then we may overestimate this required ϕ. A simple correction can be applied to our bounds to account for this issue, as shown in the following result. Given that noise arises period-by-period, we first define one-period analogues for our statistics: Denote RN movement between t and $t+1$ by $\mathsf{m}^*_{t,t+1} \equiv (\pi^*_t - \pi^*_{t+1})^2$, RN uncertainty at t by $\mathsf{u}^*_t \equiv (1-\pi^*_t)\pi^*_t$, and RN excess movement between t and $t+1$ by $X^*_{t,t+1} \equiv \mathsf{m}^*_{t,t+1} - (\mathsf{u}^*_t - \mathsf{u}^*_{t+1})$. Similar to Augenblick and Rabin (2021, section II.E), we then have the following result.

PROPOSITION 7: *Assume that the observed $\widehat{\pi}^*_t$ is measured with error with respect to the true π^*_t,*

$$\widehat{\pi}^*_t = \pi^*_t + \epsilon_t, \tag{17}$$

where $\mathbb{E}[\epsilon_t] = 0$, $\mathbb{E}[\epsilon_t\, \epsilon_{t+1}] = 0$, and $\mathbb{E}[\epsilon_{t+k}\, \pi^*_{t+k'}] = 0$ for $k, k' \in \{0, 1\}$. Denoting the observed one-period RN excess movement statistic by $\widehat{X}^*_{t,t+1}$, its relation to the true value $X^*_{t,t+1}$ in expectation is

$$\mathbb{E}\left[\widehat{X}^*_{t,t+1} - X^*_{t,t+1}\right] = 2\mathrm{Var}(\epsilon_t).$$

[23] For example, a robust finding in this literature is that the estimated SDF declines monotonically for the range of negative index returns; see, for example, Driessen, Koëter, and Wilms (2025, fig. 4) or Schreindorfer and Sichert (2025, footnote 1). As will be seen in Section IV, our empirical estimates of ϕ_j for these negative return states are somewhat higher than for positive return states, indicating that the results are unlikely to be driven by violations of Assumption 3.

We can thus subtract $2\text{Var}(\epsilon_{t,j})$ from each period's observed excess-movement statistic to obtain an unbiased true excess movement value, which can then be used in our bounds. If measurement error is positively correlated over time rather than uncorrelated, this will reduce the upward bias in measured X^*. We discuss estimation of $\text{Var}(\epsilon_t)$, and autocovariance statistics for ϵ_t, in Section IV.B.

IV. Empirical Estimation and Results

Our theory leads to bounds on the variation in RN beliefs over the index return, which we now measure in the data. We begin by describing how we map from theory to data and how we estimate microstructure noise. We then summarize the data and present our main results, along with a set of robustness checks.

A. Data and Risk-Neutral Distribution

Data: Our main source for S&P 500 index options data is the OptionMetrics database, which provides end-of-day prices for European call and put options for all strike prices and option expiration dates traded on the Chicago Board Options Exchange (CBOE). The sample runs from January 1996 through December 2018.[24] We augment these data with intraday price quotes obtained directly from the CBOE for a subset of trading days in our sample, in order to account for market microstructure noise; this additional data set is described further in Section IV.B.

We apply standard filters to remove outliers and options with poor trading liquidity from the OptionMetrics data; see Internet Appendix Section II.F.[25] Two aspects of this cleaning bear mention here. First, while our bounds apply for belief streams of arbitrary length, we follow prior literature (e.g., Christoffersen, Heston, and Jacobs, 2013; Martin, 2017) and consider options with maturity of at most one year, and streams with at least $T = 5$ observations. Second, after transforming the filtered prices to RN beliefs, we keep only conditional RN belief observations $\widetilde{\pi}^*_{t,i,j}$ for which the nonconditional beliefs satisfy $\pi^*_t(R_{T_i} = \theta_j) + \pi^*_t(R_{T_i} = \theta_{j+1}) \geqslant 5\%$, as conditional beliefs are likely to be particularly susceptible to mismeasurement when the underlying beliefs are close to zero.

[24] We use the same sample as in an earlier version of the paper (Augenblick and Lazarus, 2022), as it aligns with our intraday data for noise estimation. The 2018 cutoff precludes the volatility observed in 2020 from affecting our estimates.

[25] There are 12.4 million option prices in the raw data, and 4.3 million (for 5,537 trading dates and 991 expiration dates) after filtering. The bulk of the difference is attributable to our use of only out-of-the-money call and put strikes.

Empirical return space: For our baseline estimation, we define the return state space Θ in terms of log excess-return intervals,

$$\Theta = R^f_{0_i,T_i} \exp\big\{(-\infty, -0.2], (-0.2, -0.15], (-0.15, -0.1], \ldots, (0.1, 0.15], (0.15, 0.2], (0.2, \infty)\big\},$$

where $R^f_{0_i,T_i}$ is the gross risk-free rate from 0_i to T_i. In words, return state 1 is realized if the log excess S&P return from 0_i to T_i is less than $-0.2 \approx -20\%$, state 2 is realized if the excess return is in the 5 pp bin between -0.2 and -0.15, and so on.

More concretely, if the index value on the start date 0_i is $S_{0_i} = 100$ and the net risk-free rate is zero, then the return states will be defined so that state 1 is realized if S_{T_i} is less than (roughly) 80, state 2 if S_{T_i} is between 80 and 85, and so on. These bins are then fixed for all t until expiration, so that beliefs are over fixed events. We do not annualize returns when defining return states so that the states have the same meaning in terms of percent wealth differences regardless of the length of a given belief stream. Of course, the meaning of, for example, a -15% excess return may be different in one year than in the next. But Proposition 6 tells us that when grouping them together in estimation, our bound will still be conservative for the maximal value of ϕ_j across all panels.

Our 5 pp bin choice reflects a desire to balance (i) measurement accuracy for the RN beliefs and (ii) plausibility of our assumption of constant ϕ_j (CTI). Wider bins lead to less measurement noise but make it less likely that there are no changes in the expected SDF realization conditional on a given state θ_j relative to θ_{j+1}. We discuss alternative bin choices after presenting our main results.

For notation, we often refer to states by the right end of their return bin: $\theta_1 = -0.2$, $\theta_2 = -0.15$, and so on. The binary beliefs to be used in our tests are $\widetilde{\pi}^*_{t,i,j} \equiv \pi^*_t(R_{T_i} = \theta_j \,|\, R_{T_i} \in \{\theta_j, \theta_{j+1}\})$, so $\widetilde{\pi}^*_{t,i,j}$ corresponds to the probability that the low state j (e.g., $\theta_2 = -15\%$) will be realized, conditional on j or $j+1$ (here, conditional on an excess return between -20% and -10%). We again use the right end of the return bin for j in discussing statistics for pair (θ_j, θ_{j+1}), corresponding to the midpoint of the two bins. We report empirical estimates for all adjacent state pairs for completeness, but as discussed after Assumption 4, it is unlikely that CTI is met for the extreme state pairs (θ_1 relative to θ_2, and $\theta_9 = \theta_{J-1}$ versus $\theta_{10} = \theta_J$). Our focus is thus on the interior state pairs with low-return states $\theta_2, \ldots, \theta_8$. When we aggregate our state-by-state estimates of ϕ_j required to rationalize the data into a single average value $\overline{\boldsymbol{\phi}}$ across states, we use only these interior states.

Risk-neutral beliefs: To extract a risk-neutral distribution over the return states in Θ from the observed option cross-sections, we use standard tools from the option pricing literature. We follow Malz (2014) and fit an interpolating spline to the observed prices (in implied volatility, or IV, space). Then, using equation (13), we map fitted prices to RN beliefs by calculating numer-

ical derivatives over very small finite differences. See Internet Appendix Section II.F for details. With the RN beliefs $\pi^*_{t,i,j}$ in hand, we next calculate conditional beliefs as $\widetilde{\pi}^*_{t,i,j} = \pi^*_{t,i,j}/(\pi^*_{t,i,j} + \pi^*_{t,i,j+1})$. We use the resulting conditional RN belief streams to calculate the excess movement statistics $X^*_{i,j}$ needed to implement our bounds. Our general results in Section II restrict the expectation of $X^*_{i,j}$ conditional on state θ_j or θ_{j+1} being realized—in particular, Propositions 3 and 6 give bounds for $\widetilde{\mathbb{E}}[X^*_{i,j}]$—and we accordingly keep only those observations for which $\widetilde{\pi}^*_{T_i,i,j} = 0$ or 1 ex post; for example, if the total excess return on the market over the life of option contract i is -14%, then we keep only $X^*_{i,2}$ (θ_2 ranges from -20% to -15% return, so $\widetilde{\pi}^*_{T_i,i,2} = 0$) and $X^*_{i,3}$ ($\widetilde{\pi}^*_{T_i,i,3} = 1$).

Simplifying notation: Having clarified how the relevant empirical objects ($\widetilde{\pi}^*_{t,i,j}$, $X^*_{i,j}$) depend on the contract i and state pair j, in what follows we generally drop the cumbersome use of i, j, and $\widetilde{\cdot}$, and again write π^*_t for $\widetilde{\pi}^*_{t,i,j}$, X^* for $X^*_{i,j}$, and so on. Similarly, we often drop the "conditional" qualifier when referring to conditional RN belief statistics.

B. Noise Estimation

As in Proposition 7, we also wish to account for measurement error stemming from possible nonfundamental or microstructure noise. With $\widehat{\pi}^*_t = \pi^*_t + \epsilon_t$ as in (17), Proposition 7 tells us that we must estimate $\mathrm{Var}(\epsilon_t)$ in order to eliminate the bias in X^* arising from any such noise. We turn to a sample of high-frequency option prices to estimate this noise variance in our RN beliefs data.

Specifically, we obtain minute-by-minute price quotes on S&P index options for a subset of trading days directly from the CBOE. For each available option expiration date on each such trading day, we recalculate the RN belief distribution at the end of each minute using the same procedure as described in Section IV.A. As this requires calculating 390 sets of RN beliefs for each trading day (9:30 a.m. to 4:00 p.m.), this would be computationally infeasible if applied to our entire sample of 5,537 trading days (each of which has 11 available expiration dates on average). We thus select 30 trading days at random from our sample period and calculate intraday RN distributions for these days, yielding an intraday data set roughly twice as large as the original one.

We then use tools from the literature on microstructure noise to estimate $\mathrm{Var}(\epsilon_t)$ using these data. The intuition for this strategy is best understood by assuming temporarily that the noise ϵ_t in (17) is independent and identically distributed (i.i.d.), while the true π^*_t changes smoothly over time (e.g., Zhang, Mykland, and Aït-Sahalia, 2005). Given high-frequency option data, imagine calculating movement using the observed beliefs, $(\widehat{\pi}^*_{t+h} - \widehat{\pi}^*_t)^2$. As one decreases the time h between consecutive observations, the noise swamps the true variation: Since $(\pi^*_{t+h} - \pi^*_t)^2 \to 0$, we have $\mathbb{E}[(\widehat{\pi}^*_{t+h} - \widehat{\pi}^*_t)^2] \to 2\mathrm{Var}(\epsilon_t)$. Thus, in this example, $\mathrm{Var}(\epsilon_t)$ can be estimated using the quadratic variation in RN beliefs sampled at a high frequency.

In practice, one would expect both non-i.i.d. noise ϵ_t and jumps in the true process π_t^*. One estimator for $\text{Var}(\epsilon_t)$ that is robust to these features is the ReMeDI ("Realized moMents of Disjoint Increments") estimator proposed by Li and Linton (2022). It takes the average product of *disjoint increments* of the observed process, $(\widehat{\pi}_t^* - \widehat{\pi}_{t-h}^*)(\widehat{\pi}_t^* - \widehat{\pi}_{t+h}^*)$.[26] Intuitively, even if the true process features jumps so that $\mathbb{E}[(\pi_{t+h}^* - \pi_t^*)^2] > 0$, its nonoverlapping increments are still approximately uncorrelated. Li and Linton (2022) show that this estimator is consistent for $\text{Var}(\epsilon_t)$ for quite general noise and underlying processes, and it performs well in simulations and empirical applications.

Using this ReMeDI estimator on our minute-by-minute data, we estimate $\text{Var}(\epsilon_t) = \text{Var}(\epsilon_{t,i,j})$ separately for each combination of trading day t, expiration date T_i, and state pair j in our intraday sample. We then match the noise estimates (which are obtained for a subsample of days) to the X^* observations in our original data.[27] Finally, we subtract $2\widehat{\text{Var}}(\epsilon_{t,i,j})$ from $\widehat{X}^*_{t,t+1,i,j}$ to obtain a *noise-adjusted* estimate of one-day excess movement following Proposition 7, and we sum these noise-adjusted one-day values over the full stream to obtain noise-adjusted $X^*_{i,j}$.

We discuss the magnitude of the noise estimates in the next subsection. The ReMeDI procedure also allows for estimation of the intraday autocovariances of the noise ϵ_t. These autocovariances are estimated to be positive for small lag values, but they die out quickly and are precisely estimated near zero for noise observations more than an hour apart. This justifies the assumption in Proposition 7 that end-of-day noise observations are uncorrelated, $\mathbb{E}[\epsilon_t \epsilon_{t+1}] = 0$.

Our main results in Section IV.D use the noise-adjusted excess movement data. All standard errors and confidence intervals are based on a bootstrap procedure (detailed in Section IV.D) that accounts for the sampling uncertainty in the above noise estimation and averaging procedure.

C. Excess Movement: Descriptive Statistics and Figures

Table I summarizes the average RN excess movement $\overline{X}^*$ overall (across all interior state pairs and expiration dates) and by subsample. Excess movement is difficult to interpret without some normalization. The first two columns thus divide $\overline{X}^*$ by average initial uncertainty $\overline{u}_0^*$. As in Augenblick and Rabin (2021), this normalized statistic can be interpreted as the percent by which movement exceeds initial uncertainty and thus uncertainty resolution. These values are quite high in our data: For the noise-adjusted statistics, there is on average 123% more movement than initial uncertainty. These values decrease for re-

[26] More formally, the estimator is $\widehat{\text{Var}}(\epsilon_t) = \frac{1}{N_{\epsilon,n}} \sum_{i=2k_n}^{N_{\epsilon,n}-k_n} (\widehat{\pi}_{t_i}^* - \widehat{\pi}_{t_i-2k_n}^*)(\widehat{\pi}_{t_i}^* - \widehat{\pi}_{t_i+k_n}^*)$, where $N_{\epsilon,n}$ is the number of observations over a fixed span (in our case, one trading day) and k_n is a tuning parameter.

[27] We match these observations using the two best predictors of $\widehat{\text{Var}}(\epsilon_t)$ in our data: the state pair j and the sum $\pi_t^*(R_T = \theta_j) + \pi_t^*(R_T = \theta_{j+1})$. Further explanation and technical details can be found in Internet Appendix Section II.G.

Table I

Descriptive Statistics for Excess Movement

This table presents empirical conditional means of risk-neutral excess movement $\overline{X}^* \equiv \widehat{\mathbb{E}}[X^*_{i,j}]$, calculated over all interior state pairs $j = 2, \ldots, 8$, aside from averages by bin, which are calculated for each state pair separately. Standard errors are estimated using a block bootstrap for the normalized statistic $\overline{X}^*/\overline{\mathrm{u}}^*_0$ or $\overline{X}^*/\overline{T}$, with a block size of one month (where contracts are classified by their expiration month) and 10,000 draws.

	$\overline{X}^*/\overline{\mathrm{u}}^*_0$		$\overline{X}^*/\overline{T}$				
	Raw	Noise-Adj.	Raw	Noise-Adj.	$\overline{\mathrm{u}}^*_0$	$\overline{T}$	N (Obs.)
Overall mean:	1.89	1.23	0.0059	0.0038	0.18	56	1,809
[Bootstrapped SE]	[0.25]	[0.22]	[0.0007]	[0.0007]	[0.00]	[2]	
By return state:							
1 (−20%)	5.83	4.83	0.0049	0.0041	0.17	200	26
	[1.18]	[1.05]	[0.0013]	[0.0012]	[0.01]	[20]	
2 (−15%)	11.61	5.70	0.0180	0.0088	0.22	141	19
	[3.32]	[3.06]	[0.0046]	[0.0047]	[0.01]	[25]	
3 (−10%)	5.76	2.37	0.0151	0.0062	0.21	81	49
	[0.99]	[1.07]	[0.0025]	[0.0029]	[0.01]	[12]	
4 (−5%)	2.67	1.39	0.0088	0.0046	0.14	42	272
	[0.59]	[0.50]	[0.0016]	[0.0015]	[0.01]	[5]	
5 (0%)	0.70	0.47	0.0045	0.0030	0.23	37	700
	[0.16]	[0.14]	[0.0009]	[0.0008]	[0.00]	[2]	
6 (+5%)	1.71	1.14	0.0039	0.0026	0.11	49	567
	[0.35]	[0.34]	[0.0008]	[0.0008]	[0.01]	[3]	
7 (+10%)	3.87	2.92	0.0053	0.0040	0.18	129	144
	[1.00]	[1.03]	[0.0013]	[0.0014]	[0.01]	[9]	
8 (+15%)	5.65	5.26	0.0060	0.0056	0.21	200	58
	[1.48]	[1.48]	[0.0016]	[0.0016]	[0.01]	[11]	
9 (+20%)	3.44	2.09	0.0032	0.0020	0.22	232	36
	[0.89]	[1.27]	[0.0009]	[0.0012]	[0.01]	[9]	
By date:							
1996–2000	10.89	9.67	0.0211	0.0187	0.21	107	109
	[2.24]	[2.17]	[0.0031]	[0.0032]	[0.01]	[11]	
2001–2005	1.75	0.55	0.0042	0.0013	0.22	90	112
	[0.51]	[0.40]	[0.0011]	[0.0009]	[0.01]	[11]	
2006–2010	1.25	0.68	0.0065	0.0035	0.17	32	502
	[0.22]	[0.19]	[0.0010]	[0.0009]	[0.00]	[4]	
2011–2015	1.75	1.09	0.0050	0.0031	0.19	67	530
	[0.36]	[0.28]	[0.0010]	[0.0008]	[0.00]	[5]	
2016–2018	0.36	−0.11	0.0011	−0.0003	0.16	50	556
	[0.21]	[0.14]	[0.0006]	[0.0004]	[0.00]	[3]	
By π^*_0:							
0–0.25	1.01	0.30	0.0055	0.0017	0.09	16	185
	[0.60]	[0.55]	[0.0028]	[0.0028]	[0.01]	[2]	
0.25–0.5	1.58	0.91	0.0067	0.0039	0.23	55	883
	[0.21]	[0.17]	[0.0007]	[0.0007]	[0.00]	[3]	
0.5–0.75	2.84	2.19	0.0053	0.0041	0.23	123	284
	[0.61]	[0.58]	[0.0011]	[0.0011]	[0.00]	[7]	
0.75–1	2.54	1.88	0.0048	0.0036	0.06	31	457
	[0.95]	[0.90]	[0.0017]	[0.0016]	[0.00]	[2]	

turn states in the middle of the distribution. The early subsample has high but noisy X^* statistics (peaking during the 1998 Russian debt crisis); afterwards, excess movement decreases over time but remains high until the most recent subsample. Higher priors π_0^* correspond with greater X^*, in line with our bounds given $\phi \geqslant 1$.

The next two columns of Table I instead normalize $\overline{X}^*$ by the average contract length $\overline{T}$, so the resulting statistics can be interpreted as excess movement per day.[28] Under this normalization, there is now no clear pattern for average excess movement across bins: Longer average contract lengths tend to coincide with more excess movement, as RN beliefs bounce up and down over the length of a contract. This is inconsistent with RE, under which excess movement in subjective beliefs should not depend at all on the horizon at which uncertainty is resolved. For the splits by date and by prior, the qualitative patterns discussed above are still present, though the decrease in excess movement over time is quantitatively somewhat muted. This indicates that a share of the apparent time trend is a result of contract lengths becoming shorter after the introduction of weekly options in 2005.[29]

Comparing the raw and noise-adjusted values makes clear that despite the substantial excess movement in the noise-adjusted statistics, noise does represent a meaningful portion (about one-third) of the raw X^* data. The raw and noise-adjusted mean for $\overline{X}^*/\overline{T}$ differ by about 0.002, implying $\widehat{\mathrm{Var}}(\epsilon_t) \approx 0.002/2 = 0.001$. The standard deviation of ϵ_t is thus roughly 0.03 per day. For the return-state splits, noise tends to be lowest for returns near the center of the distribution, as is intuitive.

Next, Figure 4 provides a visual summary of the X^* statistics relative to the bounds. The blue curves describe the raw and noise-adjusted local-average X^* statistics as one varies the RN prior π_0^*; these curves are the same in both panels. As one would expect, there is very little excess movement for RN priors near zero or one, but excess movement is positive for intermediate π_0^* values for which there is greater initial uncertainty. We compare these values to the theoretical bounds under different levels of ϕ for each value of π_0^*, as shown in gray. Panel A uses the tighter bound from Proposition 6(i), with $\triangle$ estimated using local averages for the conditional expectations $\mathbb{E}[X^* \mid \theta, \pi_0^*]$ in (9) over π_0^*.[30] Panel B uses the conservative bound from the second inequality in Proposition 6(ii), so the gray lines in this panel align with the solid bound lines in Figure 3.

Across the two panels, the X^* values observed in the data exceed both sets of bounds, other than for high RN priors π_0^* and for high ϕ. As the bounds

[28] For a rough idea of the actual variation in RN beliefs given these values, consider a pair $(t, t+1)$ for which $\pi_{t+1}^* = 1 - \pi_t^*$. One-day X^* and m^* then coincide (there is no uncertainty resolution), so, for example, the noise-adjusted mean of 0.0038 corresponds to a raw change of $\sqrt{0.0038} \approx 0.06$ (or $\pi_t^* = 0.47$, $\pi_{t+1}^* = 0.53$).

[29] There remains a decrease over time in the $\overline{X}^*/\overline{T}$ statistics, possibly reflecting improvements in liquidity. Even when restricting attention to post-2000 data, though, our main results hold, with risk aversion above 100 needed to satisfy the bound.

[30] To emphasize that $\widehat{\triangle} < 0$ for $\pi_0^* < 0.5$, the figure does not cut the bounds off at zero.

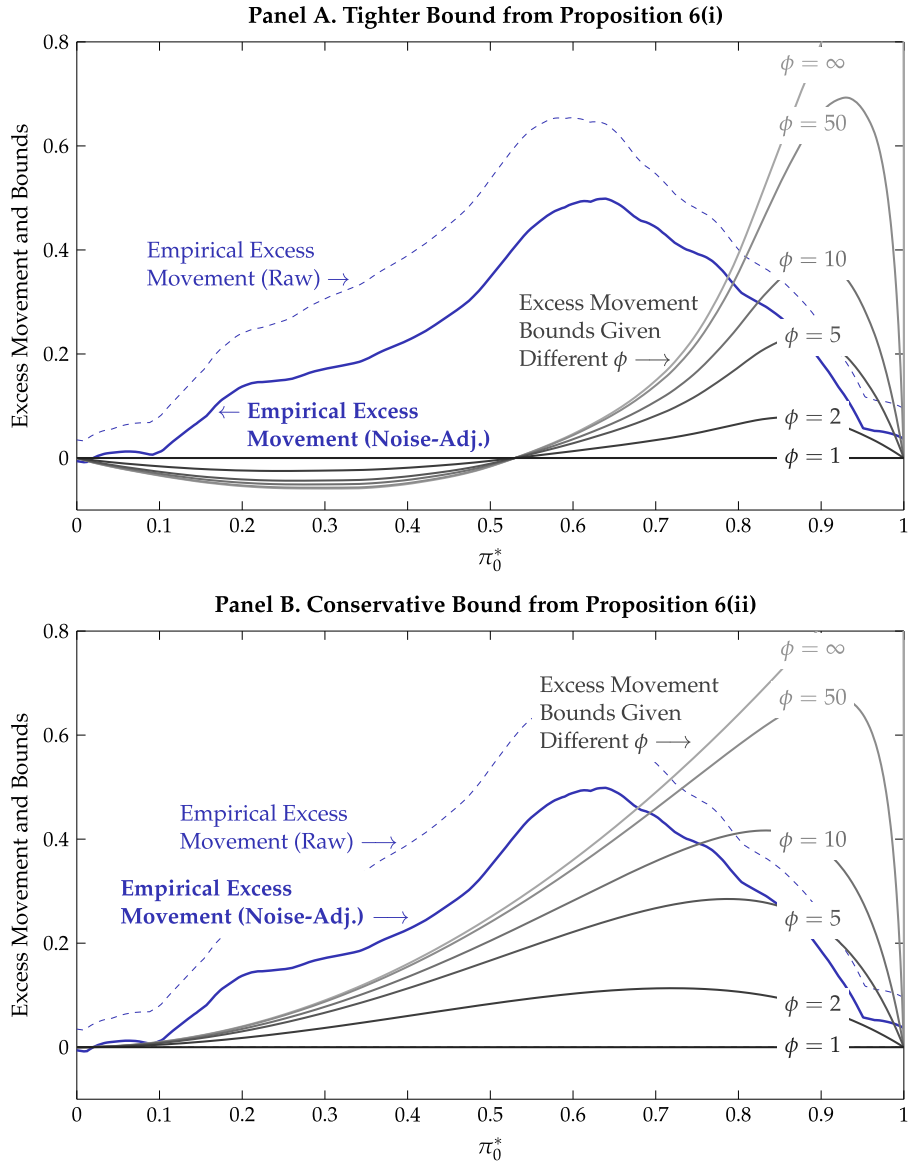


Figure 4. Excess movement versus RN prior: Data and theoretical bounds. Empirical excess movement curves are kernel-weighted averages (Epanechnikov kernel, π_0^* bandwidth of 0.08) over all interior state pairs $j = 2, \ldots, 8$. All statistics are estimated conditional means $\widetilde{\mathbb{E}}[\cdot]$ for RN beliefs $\widetilde{\pi}^*_{t,i,j}$, and theoretical curves correspond to $\phi \equiv \widetilde{\mathbb{E}}[\phi_{i,j}]$, with notation simplified for clarity. Bounds for Panel A obtain $\widehat{\Delta}$ using a kernel-weighted average over π_0^* for each of the two terms in (9), with $\widehat{\Delta}$ then plugged into the inequality in Proposition 6(i). Bounds for Panel B use only the second inequality in Proposition 6(ii). (Color figure can be viewed at wileyonlinelibrary.com)

apply for each possible π_0^*, the positive point estimates for $\mathbb{E}[X^* \mid \pi_0^*]$ clearly violate the bounds for $\pi_0^* < 0.5$, which are (at most) zero for all ϕ. And while the empirical curves are closer to the more-conservative bounds in Panel B, the noise-adjusted estimates still exceed π_0^{*2} (the bound for $\phi = \infty$) for π_0^* less than about two-thirds.

These figures do not, however, integrate over π_0^*, nor do they include any measures of statistical uncertainty necessary to make inferential statements or conduct hypothesis tests. To address these issues, we proceed to our main estimation and results.

D. Main Results

We turn now to the empirical implementation of our theoretical bounds. Given our sample of noise-adjusted excess movement statistics and corresponding RN priors, each possible value of $\overline{\phi}$ leads to a *residual* excess movement value $e_i(\overline{\phi}) = X_i^* - \text{bound}(\pi_{0,i}^*, \overline{\phi})$ for contract i. We calculate two versions of this residual corresponding to the bound in part (i) and the unconditional bound in part (ii) of Proposition 6 (equations (15) and (16), respectively):

$$\begin{aligned} e_i^{\triangle}(\overline{\phi}) &= X_i^* - \max\left\{0, \left(\pi_{0,i}^* - \frac{\pi_{0,i}^*}{\overline{\phi} + (1-\overline{\phi})\pi_{0,i}^*}\right)\widehat{\triangle}_i\right\}, \\ e_i^{\text{main}}(\overline{\phi}) &= X_i^* - \left(\pi_{0,i}^* - \frac{\pi_{0,i}^*}{\overline{\phi} + (1-\overline{\phi})\pi_{0,i}^*}\right)\pi_{0,i}^*. \end{aligned} \tag{18}$$

The first version corresponds to the tighter bound from Proposition 1, which requires a smoothed estimate of $\triangle_i$ as calculated for Figure 4. This version accounts for the estimated DGP (through $\widehat{\triangle}_i$) and thus conveys some useful preliminary information, but strictly speaking, part (i) of the proposition applies only conditional on a given $\pi_{0,i}^*$ and only under the unverifiable condition that $\text{Cov}(\pi_{0,i}, \triangle_i) = 0$. Thus, the second version, which implements the more conservative bound from Proposition 2 and which applies unconditionally, is the basis for our main set of results. Both residuals can be calculated directly using our noise-adjusted data for each possible value of $\overline{\phi}$.

We first present sample averages of these residual statistics for a range of values of $\overline{\phi}$, both for each individual state pair j and aggregated over all interior states. As these values have no natural scaling, we present them as t-statistics, $t_{\overline{e}} = \overline{e_i(\phi)}/\widehat{SE}_{\overline{e}}$, where $\overline{e_i(\phi)}$ is the sample average of $e_i(\overline{\phi})$ and $\widehat{SE}_{\overline{e}}$ is its standard error. Given the overlapping nature of option panels, we calculate standard errors with a block bootstrap. We use blocks of one month, with each block containing (i) raw excess movement statistics and priors for all contracts expiring in a given month, and (ii) noise variance estimates for any trading days in our intraday sample that fall in the same month. For each resampled data set, we use the set of $(X_i^*, \widetilde{\pi}_{0,i}^*, \{\widehat{\text{Var}}(\epsilon_{t,i})\})$ values to recalcu-

late noise-adjusted excess movement and residual values $e_i(\overline{\phi})$.[31] The bootstrap thus accounts for sampling uncertainty in all statistics used to calculate noise-adjusted X_i^* and $e_i(\overline{\phi})$. We conduct 10,000 such draws, from which we calculate standard errors as the standard deviation of $\overline{e_i(\phi)}$ across draws.

These residual t-statistics are presented in Table II for the two versions of the residual in (18), both overall and by state pair. Since $\mathbb{E}[e_i(\overline{\phi})] = 0$ under RE given a correctly specified $\overline{\phi}$, these t-statistics tell us how far the residuals are from being consistent with any hypothesized null for the SDF slope. Positive numbers correspond to the data exhibiting too much excess movement to be consistent with a given value of $\overline{\phi}$. In the first six columns, the only negative t-statistics for the tighter bound are for RN beliefs over $(\theta_j, \theta_{j+1}) = ([0\%, 5\%], [5\%, 10\%])$ (i.e., $j = 6$) for $\overline{\phi} > 10$; all other t-statistics, including the overall values, are positive (and mostly large in magnitude), indicating that no value of $\overline{\phi}$ is consistent with the degree of observed excess RN movement. In the remaining columns, the conservative bound t-statistics are somewhat smaller in magnitude but also generally positive, other than for large $\overline{\phi}$ and for returns in the middle of the distribution.

As admissible excess movement $\mathbb{E}[X_i^*]$ is monotonically increasing in the unobserved parameter $\overline{\phi}$, our main empirical exercise is to estimate the lower bound for this SDF slope such that the bound for $\mathbb{E}[X_i^*]$ is satisfied. This lower bound for $\overline{\phi}$ is estimated as the minimal value for which the average residual $\overline{e_i(\phi)}$ is zero, so that we are effectively finding the root of the function traced out in Table II. Given that the tighter bound is generally not satisfied even for $\overline{\phi} = \infty$ (and since it holds only under restrictive assumptions), we now confine attention to the conservative bound from Proposition 6(ii). The estimated $\overline{\phi}$ is the minimal SDF slope for which the amount of observed excess movement in RN beliefs can be rationalized; it is thus an index of the restrictiveness of Assumptions 2 to 4 (or, given Proposition 5, Assumptions 2 and 3 and supermartingale ϕ_t). As above, we estimate this SDF slope both for each individual state pair ($\overline{\phi} = \overline{\phi}_j$) and overall ($\overline{\phi} = \overline{\boldsymbol{\phi}}$). To make the estimates for $\overline{\phi}$ more interpretable, we also use Proposition 4 to translate them into local relative risk aversion values $\overline{\gamma} = \frac{(\overline{\phi}-1)}{0.05} = 20(\overline{\phi} - 1)$ for a representative agent who consumes the market.

Table III presents our main results. Whenever there is *no* value of $\overline{\phi}$ for which the bound for $\mathbb{E}[X_i^*]$ is satisfied—that is, from Proposition 6(iii), when we estimate $\mathbb{E}[X_i^*] > \mathbb{E}[{\pi_{0,i}^*}^2]$—we write $\overline{\phi} = \infty$. In brackets below each point estimate, we provide the lower bound of a one-sided 95% confidence interval (CI) for the parameter. To obtain these CI bounds, we start from the same bootstrap resampling procedure as described for Table II. We then search for and report the minimal value $\overline{\phi}$ such that we cannot reject that excess volatility

[31] For the residual $e_i^{\Delta}(\overline{\phi})$ corresponding to the tighter bound, we also reestimate $\widehat{\Delta}_i$ in each bootstrap draw by calculating the same local average (with respect to $\pi_{0,i}^*$) as in Figure 4, Panel A, and then evaluating it at the observed $\pi_{0,i}^*$.

Table II

Residual Excess Movement *t*-Statistics for Different $\overline{\phi}$

This table shows t-statistics for the average residuals $\overline{e_i(\phi)}$ in (18) for different values of $\overline{\phi}$. The columns for Panel A use $e_i^{\triangle}(\overline{\phi})$, with $\triangle_i$ estimated as in Panel A of Figure 4. The columns for Panel B use $e_i^{\text{main}}(\overline{\phi})$. Standard errors are estimated using a block bootstrap with block size of one month and 10,000 draws. All statistics are calculated using conditional means of noise-adjusted X^*.

	Panel A. Tighter Lower Bound *(Proposition 6(i))*						Panel B. Conservative Lower Bound *(Proposition 6(ii))*					
$\overline{\phi}$:	1	2	5	10	50	∞	1	2	5	10	50	∞
Overall t-stat.:	5.18	5.15	4.98	4.64	3.03	0.78	5.18	4.08	2.68	1.75	0.07	−1.48
By return state:												
1 (−20%)	4.06	3.69	3.04	2.57	1.94	1.72	4.06	3.58	2.79	2.22	1.47	1.21
2 (−15%)	2.13	2.13	2.13	2.13	2.14	2.14	2.13	2.03	1.94	1.90	1.87	1.86
3 (−10%)	2.23	2.23	2.23	2.23	2.23	2.23	2.23	2.03	1.86	1.79	1.72	1.70
4 (−5%)	2.78	2.78	2.77	2.77	2.76	2.76	2.78	2.51	2.28	2.18	2.09	2.06
5 (0%)	3.09	3.09	3.08	3.08	3.06	3.06	3.09	1.85	0.74	0.26	−0.19	−0.31
6 (+5%)	3.26	2.66	1.27	0.04	−1.71	−2.42	3.26	1.96	−0.12	−1.77	−5.22	−8.39
7 (+10%)	2.82	2.73	2.51	2.29	1.70	1.23	2.82	2.38	1.73	1.28	0.50	0.06
8 (+15%)	3.53	3.49	3.42	3.37	3.29	3.25	3.53	3.20	2.77	2.51	2.21	2.10
9 (+20%)	1.63	1.58	1.52	1.48	1.44	1.43	1.63	1.36	1.05	0.89	0.72	0.68

Table III
Main Estimation Results

The first column reports estimates for the minimal value of $\overline{\phi}$ satisfying the conservative bound for excess movement in Proposition 6(ii). These estimates are translated to relative risk aversion $\overline{\gamma}$ using Proposition 4, as shown in the second column. Point estimates are obtained by finding the value $\overline{\phi}$ such that $\overline{e_i^{\text{main}}(\phi)} = 0$ in (18). Confidence interval lower bounds are obtained by inverting a test for $\overline{\phi}$ using bootstrapped data; see Internet Appendix Section II.H for details. All estimates use conditional means of noise-adjusted excess movement.

	Conservative Lower Bound for:	
	SDF Slope $\overline{\phi}$	RRA $\overline{\gamma}$
Overall bound:	54.7	1,075
[95% CI lower bound]	[9.8]	[175]
By return state:		
1 (−20%)	∞	∞
	[24.2]	[464]
2 (−15%)	∞	∞
	[∞]	[∞]
3 (−10%)	∞	∞
	[∞]	[∞]
4 (−5%)	∞	∞
	[∞]	[∞]
5 (0%)	19.4	368
	[2.1]	[22]
6 (+5%)	4.8	75
	[2.2]	[24]
7 (+10%)	∞	∞
	[4.6]	[73]
8 (+15%)	∞	∞
	[∞]	[∞]
9 (+20%)	∞	∞
	[1.0]	[1]

($\overline{e_i^{\text{main}}(\phi)}$) is zero at the 5% level in the bootstrapped data. (See Internet Appendix Section II.H for further details.) We use one-sided intervals because our estimate for $\overline{\phi}$ is a lower bound for the SDF slope consistent with the data.

Starting with the overall estimates in the first row, the point estimate for the conservative lower bound for $\overline{\boldsymbol{\phi}}$ is slightly greater than 50, in line with the t-statistic of 0.07 in Table II, Panel B, for $\overline{\boldsymbol{\phi}} = 50$. This translates to an extraordinarily high estimated lower bound for $\overline{\boldsymbol{\gamma}}$ of 1,075. Values of $\overline{\boldsymbol{\phi}}$ below 9.8 (for $\overline{\boldsymbol{\gamma}}$, below 175) are rejected at the 5% level. This is our main empirical result: Under our maintained assumptions, extremely high risk aversion is needed to rationalize the large degree of excess movement in RN beliefs observed in the data.

For the individual return-state pairs, all but two of the point estimates are infinite, indicating that no amount of utility curvature (or SDF slope) can rationalize the observed excess movement such that the bounds are satisfied. For

many of the states ($j = 2, 3, 4, 8$), their CIs also have lower bounds of ∞ (or more precisely, are empty), indicating outright model rejection. Only for RN beliefs over state pairs in the middle of the distribution—that is, for $j = 5$ and 6, with return midpoints of 0% and 5%, respectively—are there finite point estimates and CIs that contain reasonable risk-aversion values of about 20. RN beliefs over these intermediate return states are thus comparatively well behaved; for all other states, there is so much mean-reverting variation in RN beliefs that the bounds are only met for implausibly large values of $\overline{\phi}_j$, if at all.

As discussed in Section III.D, the pricing kernel puzzle tends to emerge only in the range of positive returns. One can therefore focus attention on the interior states with negative returns, $j = 2$–5, for estimates that are unlikely to be affected by the possibility that $\phi < 1$. For all four of these states, the null is fully rejected. It is thus unlikely that Assumption 3 is driving our results. Similarly, by Proposition IA4, our findings cannot in general be produced solely by miscalibrated priors. This indicates that our rejection must be arising from either excessively volatile physical belief revisions or a strong violation of CTI arising from changes in the local price of risk.

E. Robustness Checks

To ensure our results are not overly sensitive to specific measurement choices, we reestimate RN excess movement and our bounds with a range of different specifications, with full results in Internet Appendix Section II.I. First, we consider a finer partition when defining index return states: Rather than our baseline of 5 pp, we use 2 pp bins. Excess movement is greater than our baseline, and the estimated bound gives $\overline{\phi} = \infty$ (CI lower bound: ∞). Second, we define states in terms of call-option delta $\frac{\partial q_{0,K}}{\partial S_0}$ as of the beginning of the option panel: State 1 corresponds to strikes with Black–Scholes delta between 1 and 0.9, state 2 between 0.9 and 0.8, and so on.[32] This similarly produces greater $\overline{X}^*$ and estimated bounds of ∞. Third, we consider alternative smoothing methods to construct RN beliefs from observed option prices. Our baseline method follows Malz (2014) and Neuberger (2012) in fitting a cubic spline interpolant through the observed prices (more specifically, their IVs). Other methods impose some degree of smoothness upfront by not requiring the fitted spline to pass through observed prices. Here, we use the approach in Bliss and Panigirtzoglou (2004), fitting a smoothing spline with roughly equal weights on goodness of fit and smoothness. We obtain slightly less excess movement (noise-adjusted $\overline{X}^*/\overline{u}_0^*$ of 117% rather than 123%) and estimated $\overline{\phi} = 26.1$ (CI bound: 7.7), corresponding to relative risk aversion of $\overline{\gamma} = 501$ (CI: 134). Finally, we extrapolate beyond the observed strikes by pasting generalized extreme value tails onto the Bliss and Panigirtzoglou (2004) RN distribution,

[32] Our estimation uses panels of varying lengths (from five to 252 trading days, with shorter streams becoming more frequent in recent years). Five pp return states are thus wide (relative to the shape of the distribution) for shorter-horizon panels. Defining states in terms of date-0 delta makes the states roughly equivalent in their share of the date-0 RN cumulative distribution function.

following Figlewski (2010); all previous tests use lognormal tails. This alternative extrapolation leads to very slightly lower X^*, with $\overline{\phi}$ estimated to be 25.4 (CI: 7.9). Overall, none of our main results are affected by any of these choices.

V. Interpreting the Results

Our results thus far appear quantitatively large: We find 123% more RN movement than uncertainty reduction, and extremely high risk aversion needed to rationalize this excess movement according to our bound. But because our statistics are based on nonlinear transformations of option prices, it may be somewhat unclear how to interpret our estimates concretely. We take up this issue here using simulations, regressions, and a quantification of excess index volatility.

A. Comparison to Simulated Option Pricing Models

A natural first question is whether leading option pricing models are capable of generating results close to those we find empirically. Many of these models feature pricing kernels that load on innovations to volatility, which can lead to violations of our CTI assumption as discussed in Section III.A. While these models are typically not fully specified equilibrium models with explicit assumptions about agents' subjective beliefs, they can be interpreted as rational models with time-varying prices of risk.[33] We therefore conduct a set of simulations to test whether such existing models are capable of generating RN excess movement statistics similar to our estimates. We consider two leading models in these simulations.

First, we consider the GARCH model proposed by Christoffersen, Heston, and Jacobs (2013, CHJ). This model features a variance-dependent pricing kernel, and it is able to match Stein's (1989) finding of apparent overreaction in long-maturity implied volatility. See Internet Appendix Section II.J for a full description of the model. We conduct 10,000 simulations of this model using the baseline parameter values estimated by CHJ. Each simulation contains a burn-in period followed by 900 nonoverlapping windows of three months (63 trading days) of simulated data. Each three-month window corresponds to a single option panel: At the beginning of a given window, there are $T = 63$ days to expiration (close to our sample mean $\overline{T}$), and we simulate spot prices every day to expiration ($t = T$), at which point the next panel starts.[34] Each day, we calculate index option prices using CHJ's solution for the spot-price characteristic function. We then construct RN beliefs over 5 pp excess-return bins as in our empirical data, from which we calculate RN statistics. In each simulation,

[33] In a previous version of this paper (Augenblick and Lazarus, 2023), we also reported simulation results for the Campbell and Cochrane (1999) habit model (see that version's Internet Appendix C.4). We omit these here, as the simulated X^* statistics were low enough that the bound estimates were one (RRA = 0) for all interior states.

[34] We use 900 option panels per simulation because this is close to the number of expiration dates in our sample.

Table IV

Excess Movement Estimates for Simulated Option Pricing Models

This table presents excess movement estimates and bounds from option pricing model simulations. All statistics are means across 10,000 simulations, with each simulation containing 900 option contracts of $T = \overline{T} = 63$ days with states defined as 5 pp excess-return ranges. See the notes to Table III for details on the bound-based estimates of $\overline{\phi}$ and $\overline{\gamma}$. The true value $\overline{\phi}_0$ is estimated by calculating the ratio of RN to physical probabilities as of date 0 for each state pair satisfying $R_T \in \{\theta_j, \theta_{j+1}\}$, and then averaging the value $\max_{\pi^*_{0,i}} \mathbb{E}[\phi_i \mid \pi^*_{0,i}]$ (from Proposition 6) across simulations. This is translated to $\overline{\gamma}_0$ using Proposition 4. See Internet Appendix Section II.J for details on the models and simulations.

			Bound Estimates		True Values	
	$\overline{X}^*/\overline{u}^*_0$	$\overline{X}^*/\overline{T}$	SDF Slope $\overline{\phi}$	RRA $\overline{\gamma}$	SDF Slope $\overline{\phi}_0$	RRA $\overline{\gamma}_0$
CHJ Model	0.11	0.0004	1.3	5.1	1.4	8.7
SVCJ Model	0.27	0.0010	1.7	13.7	1.6	12.7

we estimate average RN excess movement and estimate $\overline{\phi}$ using our bound. We then report the average estimates across simulations.

Second, we consider a stochastic volatility model in which volatility shocks are less directly tied to the index return than in CHJ. We use a model in the affine jump-diffusion class proposed by Duffie, Pan, and Singleton (2000), and in particular the version of their model with stochastic volatility with correlated jumps (SVCJ) in prices and volatility. We use estimates from Eraker, Johannes, and Polson (2003) for parameters under the physical distribution, and from Broadie, Chernov, and Johannes (2007) for parameters under the RN distribution; again see Internet Appendix Section II.J. The simulation setup and estimation otherwise parallel the description for the CHJ model above.[35]

Table IV reports results from the two sets of simulations. The CHJ model produces 11% more movement than uncertainty reduction ($\overline{X}^*/\overline{u}^*_0$), far below the 123% noise-adjusted value reported in the first row of Table I. The SVCJ model produces a significantly higher value of 27%, which is nonetheless still far below our empirical estimate.[36] The same is true for the daily excess movement statistics $\overline{X}^*/\overline{T}$, and the SVCJ value of 0.0010 is again roughly one-quarter of the empirical value of 0.0038. For the bound estimates for the SDF slope and local relative risk aversion (RRA), both models produce enough RN excess movement to obtain nonnegligible estimates for $\overline{\phi}$ and $\overline{\gamma}$.[37] Both models, however, produce estimates an order of magnitude lower than the empirical es-

[35] We solve for prices here using code adapted from Nicola Fusari (Fusari, 2017). We thank him for sharing this code.

[36] This implies that we find about 100% more excess variation in beliefs than is predicted by these models. While this may seem high, one possible point of comparison is provided by Giglio and Kelly (2018). They estimate variance ratio statistics of around two for many derivative term structures, relative to a set of affine models. A variance ratio of two implies about 100% more variation than under the null model, quite close to our estimates in a very different setting.

[37] Standard errors are very small; for example, for the SVCJ $\overline{\phi}$ estimate of 1.7, the simulation standard error is 0.001.

timates in Table III. Further, despite the violation of CTI in both models, the estimated bounds are still quite close to the models' true values of the maximal SDF slope (estimated from the ratio of RN to physical probabilities), as shown in the last two columns. The bound is slightly conservative in the CHJ model ($1.3 < 1.4$), and it only slightly overstates the true $\overline{\phi}_0$ in the SVCJ model ($1.7 > 1.6$).

Despite these models' ability to explain many empirical features of option prices and returns, they do not match the excess movement we document. Intuitively, the models are well designed to fit spot-price dynamics, features of option cross-sections (e.g., the IV smile), and average option returns. But our test exploits the fact that option prices at different strikes are highly volatile (in the time series) relative to each other (in the cross-section of strikes); that is, there is predictable mean-reverting variation in the relative value of the RN density at different strikes, even in the interior of the distribution. It is this feature of prices that existing models have a difficult time matching.

B. Predictors of RN Excess Movement

The previous simulations indicate that it is difficult to rationalize our results with well-known models featuring time-varying prices of risk, but they do not help identify what ingredients a model would need to get closer to matching the data. So for our second analysis aimed at addressing how to interpret our results concretely, we conduct a set of regressions to determine whether certain well-known variables comove strongly with observed RN excess movement. We provide details of the analysis in Internet Appendix Section II.K and describe the results here.

Table IA.IV reports results from time-series regressions in which the dependent variable is the monthly average of noise-adjusted RN excess movement $X^*_{t,t+1,i,j}$. We find that proxies for option illiquidity, trading activity, and intermediary constraints do comove positively with X^*, but only very weakly so: The monthly average option bid-ask spread, option trading volume, and a measure of net put buying pressure are all insignificant as predictors of X^*, both statistically and economically. So while they help explain a small amount of excess movement (joint $R^2 = 0.08$), these results indicate that option-market frictions are unlikely to be the main drivers of our results.

We do, however, find evidence in favor of a few economically meaningful factors as predictors of X^*. First, the average length of RN belief streams (i.e., $\overline{T_i}$ for contracts traded in the given month) strongly positively predicts excess movement, consistent with the discussion in Section IV.C. There is thus a pattern of apparent overreaction to weak signals about outcomes at distant horizons, which decreases for stronger signals at shorter horizons.[38] Second, for volatility-related predictors, we find evidence that X^* comoves strongly with

[38] In a follow-up paper (Augenblick, Lazarus, and Thaler, 2025), we investigate this pattern in much greater detail, showing that it holds as well in sports betting markets and in controlled experiments. See Section VI for further discussion.

the quantity of market uncertainty (the VIX), slightly less strongly with the price of this uncertainty (the variance risk premium, or VRP), and not at all with the volatility of risk aversion (using the risk-aversion proxy from Bekaert, Engstrom, and Xu, 2022), which can be thought of as a proxy for $\text{Var}(\phi_t)$. We thus find no evidence with this set of predictors for meaningful comovement between variation in the price of risk and RN excess movement. Third, we find that proxies for misvaluation and return reversals are significantly positively related to X^*. Together, the significant predictors explain 37% of the variation in X^*.

Taken together, these results suggest that RN excess movement is a meaningful and interpretable phenomenon: It is not significantly attributable to option-specific frictions, and it comoves strongly with variables that are intuitively related to aggregate equity valuations and excess volatility in expectations. While we find no evidence that variation in ϕ_t contributes substantially to X^*, it is of course possible that alternative models for time variation in ϕ_t are capable of generating significant excess movement, alongside alternative models of belief formation.

C. Quantifying the Effect on Overall Index Volatility

Finally, we consider what our results imply for overall index volatility. While the analysis so far documents excess movement in localized RN beliefs over individual return outcomes, this does not speak directly to the degree of excess volatility in the index as a whole. We now bridge this gap by aggregating the behavior of RN beliefs across the return distribution. Doing so provides an economically interpretable measure of excess index volatility in the spirit of Shiller (1981).

Conducting this index-level quantification requires some further structure. The index reflects an expectation with respect to the full RN distribution, $S_t = \mathbb{E}_t^*[S_T] = \int S_T \, d\mathbb{P}_t^*(S_T)$, making it sensitive to time variation in the aggregate price or quantity of risk.[39] Our binarized-belief tests, meanwhile, require only the relatively weak assumption of a constant *local* price of risk over small index changes. To extend this logic to the index, we restrict attention to the center of the return distribution: for date t and option expiration T, we construct an alternative, *tail-free*, index value $S_{t,\text{alt}} = \mathbb{E}_t^*[S_T \,|\, \underline{S} \leqslant S_T \leqslant \overline{S}]$, where $\underline{S}$ and $\overline{S}$ are lower and upper cutoffs for the terminal index value (corresponding to net excess returns of $\underline{\theta}$ and $\overline{\theta}$).[40] Removing tail exposure makes $S_{t,\text{alt}}$ less sensitive to changes in jump or disaster risk than the full index S_t. It is therefore plausible to assume a small and approximately constant price of risk

[39] In defining $S_t = \mathbb{E}_t^*[S_T]$, we ignore the risk-free rate and dividends; the value S_t is in effect the forward price for a dividend-stripped index. Given the horizon ($\overline{T} < 3$ months on average) and use of daily changes, dividend uncertainty and risk-free rate variation are unlikely to play a major role in this analysis.

[40] Equivalently, $S_{t,\text{alt}}$ is the strike on a forward contract with no upfront cost and payoff $(S_T - S_{t,\text{alt}})\mathbb{1}\{\underline{S} \leqslant S_T \leqslant \overline{S}\}$.

Table V
Option-Implied Excess Index Variance

This table presents average excess index movement normalized by average initial uncertainty (implied variance). We construct the index $S_{t,\text{alt}}$ (and implied variance) for forward horizon T using the option-based RN distribution confined to the range $[\underline{\theta}, \overline{\theta}]$. The first column uses the full distribution. For each T, movement is realized variance for $S_{t,\text{alt}}$ from 0 to T. Standard errors use a block bootstrap for the statistic $\overline{X}(S_{\text{alt}})/\overline{\mathsf{u}}_0(S_{\text{alt}})$, with a block size of one month and 10,000 draws. See Internet Appendix Section II.L for details.

		Net Excess-Return Range $[\underline{\theta}, \overline{\theta}]$ Used to Define the Tail-Free Index $S_{t,\text{alt}}$			
	Full ($S_{t,\text{alt}} = S_t$)	$[-10\%, 10\%]$	$[-8\%, 8\%]$	$[-6\%, 6\%]$	$[-4\%, 4\%]$
$\overline{X}(S_{\text{alt}})/\overline{\mathsf{u}}_0(S_{\text{alt}})$	−0.14	0.15	0.19	0.33	0.45
[Bootstrapped SE]	[0.03]	[0.04]	[0.04]	[0.06]	[0.07]

for $S_{t,\text{alt}}$, such that changes in $S_{t,\text{alt}}$ reflect variation in the subjective distribution $\mathbb{P}_t(S_T \,|\underline{S} \leqslant S_T \leqslant \overline{S})$. Given this, we can extend the martingale restriction from Lemma 1 to $S_{t,\text{alt}}$ and test whether its movement is consistent with its ex ante variance.

We construct $S_{t,\text{alt}}$ and its statistics directly from the option-implied RN distribution over index outcomes S_T, effectively aggregating the beliefs studied in our main tests into the relevant index-level quantities. Specifically, we consider a range of different tail cutoffs—from ±10% to ±4%, as well as a no-cutoff case ($\pm\infty$) for comparison—and for each cutoff, we use the option-implied histogram over 2 pp return bins to compute the no-arbitrage price $S_{t,\text{alt}}$ for each date t and expiration T.[41] Each option panel gives us an implied index stream $S_{\text{alt}} = \{S_{t,\text{alt}}\}$ from 0 to T, for which we define movement $\mathsf{m}(S_{\text{alt}}) = \sum_{t=0}^{T-1}(S_{t+1,\text{alt}} - S_{t,\text{alt}})^2$, ex ante implied RN variance $\mathsf{u}_0(S_{\text{alt}}) = \sum_{\underline{S} \leqslant s \leqslant \overline{S}} \mathbb{P}_0^*(S_T = s|\underline{S} \leqslant S_T \leqslant \overline{S}) \cdot (s - S_{0,\text{alt}})^2$, and excess movement $X(S_{\text{alt}}) = \mathsf{m}(S_{\text{alt}}) - \mathsf{u}_0(S_{\text{alt}})$. We assume formally that $S_{t,\text{alt}}$ is a square-integrable martingale under RE, and the resulting restriction is $\mathbb{E}[X(S_{\text{alt}})] = 0$. We test this by measuring the sample mean $\overline{X}(S_{\text{alt}})$ using our S&P options data.

Table V reports the results for different choices of the central return range $[\underline{\theta}, \overline{\theta}]$ used to construct the tail-free index $S_{t,\text{alt}}$. As in Table I, we normalize $\overline{X}(S_{\text{alt}})$ by the mean $\overline{\mathsf{u}}_0(S_{\text{alt}})$ to obtain a measure interpretable as the percent excess movement in the index. The first column shows that when we use the full distribution without excluding the tails ($S_{t,\text{alt}} = S_t$), we estimate 14% too *little* variance for this unhedged index. This is to be expected in light of the

[41] That is, $S_{t,\text{alt}} = \sum_{\underline{S} \leqslant s \leqslant \overline{S}} \mathbb{P}_t^*(S_T = s|\underline{S} \leqslant S_T \leqslant \overline{S}) \cdot s$, where s is the midpoint of the associated return bin; see Internet Appendix Section II.L. We use 2 pp bins so that the midpoint approximates the expected value. To equalize the scale of movement and uncertainty across streams, we normalize $S_{0,\text{alt}} = \frac{1}{R^f_{0,T}}$. All statistics are thus in excess-return space.

well-known VRP: For the index as a whole, RN implied variance on average exceeds realized variance.[42]

The remaining columns of Table V show that once the tails are excluded, we estimate significantly too much movement in the index, ranging from 15% to 45% as we narrow the window. The pattern across columns is intuitive. The tails are evidently the main contributor to the VRP,[43] so progressively removing them strips out the negative VRP component and allows the positive excess movement documented in our main results to dominate. The narrowest window most cleanly isolates belief-driven excess movement, but it also comes close to simply restating the local results from the center return states in Tables I to III. Balancing these considerations by focusing on the midpoint across the four rightmost columns, we estimate that the tail-hedged S&P index exhibits roughly 30% excess movement, or 30% more variance than it would under RE.

Translating this into a statement for the full unhedged index requires an assumption about excess variance in the tails, given the confounding effect of the VRP. All else equal, a larger VRP pushes $\overline{X}$ downward by increasing ex ante risk-neutral variance; this reflects compensation for bearing variance risk, and not the degree of *realized* variance driven by excess movement in physical beliefs (our object of interest). Since the VRP is concentrated in the tails, we cannot separately identify the two components for the full index. We therefore consider two benchmark assumptions.

If we assume that excess movement in physical beliefs is constant throughout the return distribution—so that the entire decrease in $\overline{X}/\bar{u}_0$ as we widen the window reflects the growing VRP contribution—then we can apply our tail-hedged estimate directly, implying roughly 30% excess variance for the index overall. We view this as reasonable given that Tables I to III show, if anything, greater excess movement in local RN beliefs outside the $\pm 10\%$ range than in the center. But as a more conservative benchmark, we can assume that the tails exhibit no belief-driven excess movement at all. In this case, the estimate falls to roughly 20% excess variance in S_t, since the center accounts for roughly two-thirds of overall index variance. While both estimates are approximations, together they suggest that the S&P 500 exhibits roughly 20% to 30% more variance than it would under RE. Our results on RN beliefs therefore imply meaningful excess volatility in the overall index.

[42] Note that $X(S_{\text{alt}})$ is equivalent to the payoff on a variance swap contract: m is equal to realized variance, and u_0 is the swap strike (implied variance). We construct this payoff $X(S_{\text{alt}})$ using the RN distribution, and it is reassuring that this approach yields evidence consistent with past work directly measuring the VRP (e.g., Carr and Wu, 2009).

[43] This result is similar in spirit to the findings of Beason and Schreindorfer (2022). They document that the left tail of the distribution accounts for the bulk of the equity premium, while we find that the tails similarly account for the bulk of the VRP. Our main focus here is on the excess index variance in the center of the distribution.

VI. Conclusion

We derive new bounds on the admissible rational variation in risk-neutral beliefs as expressed in asset prices. Unlike in much of the previous literature, these results do not require any restrictions on the DGP, and they allow for meaningful time variation in discount rates. Further, by using asset prices, we do not require direct measures of physical beliefs over future outcomes, and our bounds exploit intertemporal consistency requirements of rational beliefs without the need for the econometrician to know what agents' beliefs "should" be under RE.

When taken to the data using risk-neutral beliefs over the return on the S&P 500 index, we find that these RN beliefs are so volatile that our bounds are routinely violated. This implies a violation of our joint assumptions. We provide evidence suggesting that RE violations by the marginal investor are likely to be at least partly responsible for this violation, though we remain open to alternative forms of variation in the local price of risk playing a part. Our framework nonetheless allows us to rule out other explanations for the excess volatility we document, providing a meaningful improvement in understanding the restrictions implied by the observed volatility in prices. Further, our results are not well explained by leading option pricing models, and we show that they imply meaningful excess variation in the underlying index.

In follow-up work (Augenblick, Lazarus, and Thaler, 2025), we more directly consider what alternative models might better fit the data. We show that a model featuring overreaction to weak signals, which weakens and then reverses in response to sufficiently strong signals—as raised briefly in Section V.B—explains empirical patterns not only in option markets, but also in sports betting markets and in a range of controlled experiments. The fact that we find positive unconditional excess movement in the current paper suggests, according to this explanation, that day-to-day information is fairly uninformative about future index outcomes at the aggregate level. Applying such a model quantitatively to option prices, however, remains to be done.

We believe that there are numerous feasible ways to make additional progress in identifying the specific causes of our bound violations. Conducting additional tests on the empirical correlates of excess movement, for example, may provide useful information, as would generalizing our analysis to alternative asset classes. Finally, detailed data on changes in individual portfolios could allow for tests on the rationality of individual beliefs, which would help distinguish between micro and macro explanations for the observed excess movement in RN beliefs.

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Appendix: Proofs of Main Results

This appendix contains proofs for the main results in Sections I and II. The remaining proofs can be found in the Internet Appendix, along with additional technical materials and discussions.

A.1. Proofs for Section I

See Internet Appendix Section I.A for a rederivation of Lemma 1. Some preliminaries are useful before proceeding to the main proofs. Start by defining the *RN measure* for this section as

$$\mathbb{P}^*(H_T) \equiv \begin{cases} \mathbb{P}(H_T)\frac{\pi_0^*}{\pi_0} & \text{if } \pi_T(H_T) = 1 \\ \mathbb{P}(H_T)\frac{1-\pi_0^*}{1-\pi_0} & \text{if } \pi_T(H_T) = 0, \end{cases} \tag{A1}$$

where $\mathbb{P}(H_T)$ is the probability of observing history H_T under *DGP*. This $\mathbb{P}^*$ is a change of measure that adjusts the frequency of each signal history such that a person with RN beliefs has RE. (As shown in Lemma A3, (A1) is consistent with the usual definition in a general setting.)

LEMMA A1: *Define $\mathbb{E}^*[\cdot]$ to be the expectation under $\mathbb{P}^*$. Under Assumption 1 (RE):*

(i) For any H_T and θ, $\mathbb{P}^*(H_T|\theta) = \mathbb{P}(H_T|\theta)$.
(ii) For any H_t, $\pi_t^*(H_t) = \mathbb{E}^*[\pi_{t+1}^*(H_{t+1})|H_t]$.
(iii) For any DGP, $\mathbb{E}^*[X^*] = 0$.

PROOF OF LEMMA A1: For the physical measure,

$$\begin{aligned} \mathbb{P}(H_T) &= \mathbb{P}(\theta = 1) \cdot \mathbb{P}(H_T|\theta = 1) + \mathbb{P}(\theta = 0) \cdot \mathbb{P}(H_T|\theta = 0) \\ &= \pi_0 \cdot \mathbb{P}(H_T|\theta = 1) + (1 - \pi_0) \cdot \mathbb{P}(H_T|\theta = 0), \end{aligned} \tag{A2}$$

where the second line uses $\pi_0 = \mathbb{E}[\theta] = \mathbb{P}(\theta = 1)$ by RE. For the RN measure, (A1) and (A2) give

$$\begin{aligned} \mathbb{P}^*(H_T) &= \tfrac{\pi_0^*}{\pi_0} \cdot \pi_0 \cdot \mathbb{P}(H_T|\theta = 1) + \tfrac{1-\pi_0^*}{1-\pi_0} \cdot (1 - \pi_0) \cdot \mathbb{P}(H_T|\theta = 0) \\ &= \pi_0^* \cdot \mathbb{P}(H_T|\theta = 1) + (1 - \pi_0^*) \cdot \mathbb{P}(H_T|\theta = 0). \end{aligned} \tag{A3}$$

For any H_T such that $\pi_T = 1$, (A1) gives $\mathbb{P}^*(H_T) = \frac{\pi_0^*}{\pi_0}\mathbb{P}(H_T)$, so $\mathbb{P}^*(\theta = 1) = \frac{\pi_0^*}{\pi_0}\mathbb{P}(\theta = 1)$. Thus, $\mathbb{P}^*(H_T|\theta = 1) = \mathbb{P}(H_T|\theta = 1)$. Similarly, $\mathbb{P}^*(H_T|\theta = 0) = \mathbb{P}(H_T|\theta = 0)$, proving part (i).

Given this, (A3) becomes $\mathbb{P}^*(H_T) = \pi_0^* \cdot \mathbb{P}^*(H_T|\theta = 1) + (1 - \pi_0^*) \cdot \mathbb{P}^*(H_T|\theta = 0)$. Summing over all H_T for which $\theta = 1$ gives $\pi_0^* = \mathbb{P}^*(\theta = 1)$, so $\mathbb{P}^*$ is a valid distribution for which the law of iterated expectations holds. Part (ii) then follows from $\mathbb{P}^*(\theta = 1) = \mathbb{E}^*[\theta] = \mathbb{E}^*[\pi_T] = \mathbb{E}^*[\pi_T^*]$.

Finally, given (ii), the same proof for Lemma 1 (see Internet Appendix Section I.A) gives $\mathbb{E}^*[X^*] = 0$, as expected RN movement under the RN measure must equal RN initial uncertainty. This proves part (iii). □

Part (i) says that compared to the physical measure, the RN measure places a higher likelihood on all signal histories resolving in state 1, but does so proportionally, so that likelihoods of signal histories *conditional on state 1* do not change. This implies that conditional expectations under the two respective measures are equal. Therefore, $\mathbb{E}^*[X^*|\theta] = \mathbb{E}[X^*|\theta]$ (see Lemma A4 below for the analogue of this result in the general asset-pricing setting). This implies

$$\begin{aligned}\mathbb{E}^*[X^*] &= \pi_0^* \cdot \mathbb{E}^*[X^*|\theta = 1] + (1 - \pi_0^*) \cdot \mathbb{E}^*[X^*|\theta = 0] \\ &= \pi_0^* \cdot \mathbb{E}[X^*|\theta = 1] + (1 - \pi_0^*) \cdot \mathbb{E}[X^*|\theta = 0] = 0, \end{aligned} \tag{A4}$$

where the last equality applies part (iii) of the lemma. For $\mathbb{E}[X^*]$, it is useful to similarly write

$$\mathbb{E}[X^*] = \pi_0 \cdot \mathbb{E}[X^*|\theta = 1] + (1 - \pi_0) \cdot \mathbb{E}[X^*|\theta = 0]. \tag{A5}$$

Now recall that $\triangle \equiv \mathbb{E}[X^*|\theta = 0] - \mathbb{E}[X^*|\theta = 1]$. It will be useful to bound $\triangle$ as follows.

LEMMA A2: *For any DGP,* $\triangle \leqslant \pi_0^*$.

PROOF OF LEMMA A2: First, we can write $\triangle \equiv \mathbb{E}^*[X^*|\theta = 0] - \mathbb{E}^*[X^*|\theta = 1]$ from Lemma A1, or

$$\triangle = \mathbb{E}^*[\mathsf{m}^*|\theta = 0] - \mathsf{u}_0^* - (\mathbb{E}^*[\mathsf{m}^*|\theta = 1] - \mathsf{u}_0^*) = \mathbb{E}^*[\mathsf{m}^*|\theta = 0] - \mathbb{E}^*[\mathsf{m}^*|\theta = 1]. \tag{A6}$$

Further, using (A4),

$$\begin{aligned} 0 &= \pi_0^* \cdot \mathbb{E}[X^*|\theta = 1] + (1 - \pi_0^*) \cdot \mathbb{E}[X^*|\theta = 0] \\ &= \pi_0^* \cdot (\mathbb{E}[\mathsf{m}^*|\theta = 1] - \mathsf{u}_0^*) + (1 - \pi_0^*) \cdot (\mathbb{E}[\mathsf{m}^*|\theta = 0] - \mathsf{u}_0^*), \end{aligned}$$

so from the definition of u_0^*,

$$\pi_0^* \cdot \mathbb{E}[\mathsf{m}^*|\theta = 1] + (1 - \pi_0^*) \cdot \mathbb{E}[\mathsf{m}^*|\theta = 0] = \pi_0^*(1 - \pi_0^*). \tag{A7}$$

Solving for $\mathbb{E}[\mathsf{m}^*|\theta = 0]$ gives $\mathbb{E}[\mathsf{m}^*|\theta = 0] = \pi_0^* - \frac{\pi_0^*}{1-\pi_0^*} \cdot \mathbb{E}[\mathsf{m}^*|\theta = 1]$. Using this in (A6), we have

$$\triangle = \pi_0^* - \frac{\pi_0^*}{1 - \pi_0^*} \cdot \mathbb{E}[\mathsf{m}^*|\theta = 1] - \mathbb{E}^*[\mathsf{m}^*|\theta = 1] = \pi_0^* - \frac{1}{1 - \pi_0^*} \cdot \mathbb{E}[\mathsf{m}^*|\theta = 1]. \tag{A8}$$

Given that $\frac{1}{1-\pi_0^*} \geqslant 0$ and $\mathbb{E}[\mathsf{m}^*|\theta = 1] \geqslant 0$, $\triangle$ is bounded above by π_0^*. □

We can now proceed to our main proofs.

PROOF OF PROPOSITION 1: Start from equation (A5) and apply equation (A4):

$$\begin{aligned}\mathbb{E}[X^*] &= \pi_0 \cdot \mathbb{E}[X^*|\theta=1] + (1-\pi_0)\cdot \mathbb{E}[X^*|\theta=0] - 0 \\ &= \pi_0 \cdot \mathbb{E}[X^*|\theta=1] + (1-\pi_0)\cdot \mathbb{E}[X^*|\theta=0] - (\pi_0^* \cdot \mathbb{E}[X^*|\theta=1] + (1-\pi_0^*)\cdot \mathbb{E}[X^*|\theta=0]) \\ &= (\pi_0^* - \pi_0)(\mathbb{E}[X^*|\theta=0] - \mathbb{E}[X^*|\theta=1]) = (\pi_0^* - \pi_0)\triangle, \end{aligned} \tag{A9}$$

as stated. Then the second equality holds using equation (8) and the definition of $\triangle$. □

PROOF OF PROPOSITION 2: From Lemma A2 above, we have $\triangle \leqslant \pi_0^*$. Further, equation (8) implies

$$\pi_0^* - \pi_0 = \pi_0^* - \frac{\pi_0^*}{\pi_0^* + \phi(1-\pi_0^*)} = \pi_0^*\left(1 - \frac{1}{\pi_0^* + \phi(1-\pi_0^*)}\right) \geqslant 0, \tag{A10}$$

where the last inequality uses $\pi_0^* + \phi(1-\pi_0^*) \geqslant 1$ since $\phi \geqslant 1$. Using these two inequalities in the expression for $\mathbb{E}[X^*]$ in (A9) gives

$$\mathbb{E}[X^*] = (\pi_0^* - \pi_0)\triangle \leqslant (\pi_0^* - \pi_0)\pi_0^*. \tag{A11}$$

Plugging in the expression for $\pi_0^* - \pi_0$ in (A10) then gives equation (10). □

PROOF OF COROLLARY 1: This is an immediate implication of (A11) and $\pi_0 \geqslant 0$. □

A.2. Proofs for Section II

Again see Internet Appendix Section I.A for a derivation of equation (13). A pair of lemmas will also be useful in proving this section's results. As usual, assume throughout that Assumptions 2 to 4 hold.

LEMMA A3: *For some return-state pair* (θ_j, θ_{j+1})*, with* $\widetilde{\mathbb{P}} \equiv \mathbb{P}(\cdot \,|\, R_T \in \{\theta_j, \theta_{j+1}\})$*, define a new pseudo-risk-neutral measure* $\widetilde{\mathbb{P}}^\diamond$ *by*

$$\left.\frac{d\widetilde{\mathbb{P}}^\diamond}{d\widetilde{\mathbb{P}}}\right|_{H_t} = \frac{\widetilde{\pi}^*_{t,j}}{\widetilde{\pi}_{t,j}}\,\mathbb{1}\{R_T = \theta_j\} + \frac{1-\widetilde{\pi}^*_{t,j}}{1-\widetilde{\pi}_{t,j}}\,\mathbb{1}\{R_T = \theta_{j+1}\}. \tag{A12}$$

Denote the conditional expectation under $\widetilde{\mathbb{P}}^\diamond$ *by* $\widetilde{\mathbb{E}}^\diamond_t[\,\cdot\,]$*. If CTI holds for the return-state pair* (θ_j, θ_{j+1})*, and* $\mathbb{P}_t(R_T \in \{\theta_j, \theta_{j+1}\}) > 0$*, we have that* $\widetilde{\mathbb{P}}^\diamond$ *serves as a martingale measure for the risk-neutral belief in the sense that*

$$\widetilde{\pi}^*_{t,j} = \widetilde{\mathbb{E}}^\diamond_t[\widetilde{\pi}^*_{t+1,j}]. \tag{A13}$$

We conclude from Lemma 1 that

$$\widetilde{\mathbb{E}}^\diamond_0[X^*_j] = 0. \tag{A14}$$

PROOF OF LEMMA A3: Following the discussion after equation (14), we have that

$$\frac{\widetilde{\pi}^*_{t,j}}{\widetilde{\pi}_{t,j}} = \frac{\phi_j}{1+\widetilde{\pi}_{t,j}(\phi_j-1)}, \qquad \frac{1-\widetilde{\pi}^*_{t,j}}{1-\widetilde{\pi}_{t,j}} = \frac{1}{1+\widetilde{\pi}_{t,j}(\phi_j-1)}. \tag{A15}$$

Note, therefore, that $\widetilde{\mathbb{P}}^\diamond$ is absolutely continuous with respect to $\widetilde{\mathbb{P}}$.

Recall that $H_t = \sigma(s_\tau, 0 \leqslant \tau \leqslant t)$, where $\sigma(s_\tau, 0 \leqslant \tau \leqslant t)$ is the σ-algebra generated by $\{s_t\}$, with signals $s_t \in \mathcal{S}$. Denote $N_\mathcal{S} \equiv |\mathcal{S}|$, so $s_t \in \{s_1, s_2, \ldots, s_{N_\mathcal{S}}\}$, and further denote $\mathfrak{p}_{t,k} \equiv \widetilde{\mathbb{P}}_t(s_{t+1} = s_k)$, $\varrho_{t,k} \equiv \widetilde{\mathbb{P}}_t(R_T = \theta_j \mid s_{t+1} = s_k)$, and $\varrho^*_{t,k} \equiv \mathbb{P}^*_t(R_T = \theta_j \mid s_{t+1} = s_k, R_T \in \{\theta_j, \theta_{j+1}\})$, so $\widetilde{\pi}_{t+1,j} = \varrho_{t,k}$ and $\widetilde{\pi}^*_{t+1,j} = \varrho^*_{t,k}$ if $s_{t+1} = s_k$. Combining (A12), (A15), and these definitions yields

$$\begin{aligned}
\widetilde{\mathbb{E}}^\diamond_t[\widetilde{\pi}^*_{t+1,j}] &= \frac{\widetilde{\pi}^*_{t,j}}{\widetilde{\pi}_{t,j}} \sum_{k=1}^{N_\mathcal{S}} \mathfrak{p}_{t,k}\, \varrho^*_{t,k}\, \widetilde{\mathbb{E}}_t\big[\mathbb{1}\{R_T = \theta_j\} \mid s_{t+1} = s_k\big] \\
&\quad + \frac{1-\widetilde{\pi}^*_{t,j}}{1-\widetilde{\pi}_{t,j}} \sum_{k=1}^{N_\mathcal{S}} \mathfrak{p}_{t,k}\, \varrho^*_{t,k}\, \widetilde{\mathbb{E}}_t\big[\mathbb{1}\{R_T = \theta_{j+1}\} \mid s_{t+1} = s_k\big] \\
&= \frac{\phi_j}{1+\widetilde{\pi}_{t,j}(\phi_j-1)} \sum_{k=1}^{N_\mathcal{S}} \mathfrak{p}_{t,k}\, \frac{\phi_j \varrho_{t,k}}{1+\varrho_{t,k}(\phi_j-1)}\, \varrho_{t,k} \\
&\quad + \frac{1}{1+\widetilde{\pi}_{t,j}(\phi_j-1)} \sum_{k=1}^{N_\mathcal{S}} \mathfrak{p}_{t,k}\, \frac{\phi_j \varrho_{t,k}}{1+\varrho_{t,k}(\phi_j-1)}\, (1-\varrho_{t,k}) \\
&= \frac{\phi_j}{1+\widetilde{\pi}_{t,j}(\phi_j-1)} \sum_{k=1}^{N_\mathcal{S}} \mathfrak{p}_{t,k}\, \frac{\varrho_{t,k}\big(1+\varrho_{t,k}(\phi_j-1)\big)}{1+\varrho_{t,k}(\phi_j-1)} \\
&= \frac{\phi_j}{1+\widetilde{\pi}_{t,j}(\phi_j-1)} \sum_{k=1}^{N_\mathcal{S}} \mathfrak{p}_{t,k}\, \varrho_{t,k} = \frac{\phi_j \widetilde{\pi}_{t,j}}{1+\widetilde{\pi}_{t,j}(\phi_j-1)} = \widetilde{\pi}^*_{t,j},
\end{aligned}$$

where the second-to-last equality uses $\widetilde{\pi}_{t,j} = \widetilde{\mathbb{E}}_t[\widetilde{\pi}_{t+1,j}]$, as can be seen from the law of iterated expectations (LIE) given that $\widetilde{\pi}_{t,j} = \mathbb{E}_t[\mathbb{1}\{R_T = \theta_j\} \mid R_T \in \{\theta_j, \theta_{j+1}\}] = \widetilde{\mathbb{E}}_t[\mathbb{1}\{R_T = \theta_j\}] = \widetilde{\mathbb{E}}_t[\widetilde{\mathbb{E}}_{t+1}[\mathbb{1}\{R_T = \theta_j\}]] = \widetilde{\mathbb{E}}_t[\widetilde{\pi}_{t+1,j}]$, and the last equality above again uses (A15). Then $\widetilde{\mathbb{E}}^\diamond_0[X^*_j] = 0$ follows immediately from the proof of Lemma 1. □

LEMMA A4: *For any return-state pair (θ_j, θ_{j+1}) meeting CTI, for $j' = j, j+1$, RN movement must satisfy*

$$\widetilde{\mathbb{E}}^\diamond_0[\mathsf{m}^*_j \mid R_T = \theta_{j'}] = \widetilde{\mathbb{E}}_0[\mathsf{m}^*_j \mid R_T = \theta_{j'}]. \tag{A16}$$

PROOF OF LEMMA A4: The stream of RN beliefs is $\boldsymbol{\pi}^*_j$, and denote some arbitrary realization for that path by $\mathfrak{b}_j$. For any $\mathfrak{b}_j$ such that $\widetilde{\pi}^*_{T,j} = 1$ (i.e., $R_T = \theta_j$),

the definition of $\widetilde{\mathbb{P}}^{\diamond}$ in (A12) gives

$$\widetilde{\mathbb{P}}_0^{\diamond}(\boldsymbol{\pi}_j^* = \mathfrak{b}_j) = \frac{\widetilde{\pi}_{0,j}^*}{\widetilde{\pi}_{0,j}} \widetilde{\mathbb{P}}(\boldsymbol{\pi}_j^* = \mathfrak{b}_j), \tag{A17}$$

and further $\widetilde{\mathbb{P}}_0^{\diamond}(R_T = \theta_j) = (\widetilde{\pi}_{0,j}^*/\widetilde{\pi}_{0,j})\widetilde{\mathbb{P}}_0(R_T = \theta_j)$ trivially. Combining these two equations yields $\widetilde{\mathbb{P}}_0^{\diamond}(\boldsymbol{\pi}_j^* = \mathfrak{b}_j \mid R_T = \theta_j) = \widetilde{\mathbb{P}}_0(\boldsymbol{\pi}_j^* = \mathfrak{b}_j \mid R_T = \theta_j)$. (Intuitively, all paths ending in $\widetilde{\pi}_{T,j}^* = 1$ receive the same change of measure under $\widetilde{\mathbb{P}}^{\diamond}$ relative to $\widetilde{\mathbb{P}}$, so probabilities conditional on $R_T = \theta_j$ are preserved, and similarly for $R_T = \theta_{j+1}$, as was the case for the simpler version in (A1).) Thus,

$$\begin{aligned}\widetilde{\mathbb{E}}_0^{\diamond}[\mathsf{m}_j^* \mid R_T = \theta_j] &= \sum_{\mathfrak{b}_j\,:\,\widetilde{\pi}_{T,j}^*=1} \mathsf{m}_j^*(\mathfrak{b}_j)\,\widetilde{\mathbb{P}}_0^{\diamond}\big(\boldsymbol{\pi}_j^* = \mathfrak{b}_j \;\big|\; R_T = \theta_j\big) \\ &= \sum_{\mathfrak{b}_j\,:\,\widetilde{\pi}_{T,j}^*=1} \mathsf{m}_j^*(\mathfrak{b}_j)\,\widetilde{\mathbb{P}}_0\big(\boldsymbol{\pi}_j^* = \mathfrak{b}_j \;\big|\; R_T = \theta_j\big) = \widetilde{\mathbb{E}}_0[\mathsf{m}_j^* \mid R_T = \theta_j].\end{aligned}$$

The same applies for $R_T = \theta_{j+1}$: For any $\mathfrak{b}_j$ such that $\widetilde{\pi}_{T,j}^* = 0$, (A17) now becomes $\widetilde{\mathbb{P}}_0^{\diamond}(\boldsymbol{\pi}_j^* = \mathfrak{b}_j) = (1 - \widetilde{\pi}_{0,j}^*)/(1 - \widetilde{\pi}_{0,j})\widetilde{\mathbb{P}}(\boldsymbol{\pi}_j^* = \mathfrak{b}_j)$. Further, $\widetilde{\mathbb{P}}_0^{\diamond}(R_T = \theta_{j+1}) = (1 - \widetilde{\pi}_{0,j}^*)/(1 - \widetilde{\pi}_{0,j})\widetilde{\mathbb{P}}_0(R_T = \theta_{j+1})$, so again $\widetilde{\mathbb{P}}_0^{\diamond}(\boldsymbol{\pi}_j^* = \mathfrak{b}_j \mid R_T = \theta_{j+1}) = \widetilde{\mathbb{P}}_0(\boldsymbol{\pi}_j^* = \mathfrak{b}_j \mid R_T = \theta_{j+1})$. Thus, $\widetilde{\mathbb{E}}_0^{\diamond}[\mathsf{m}_j^* \mid R_T = \theta_{j+1}] = \widetilde{\mathbb{E}}_0[\mathsf{m}_j^* \mid R_T = \theta_{j+1}]$. □

Note that definition (A12) aligns with the measure defined in (A1), so Lemmas A3 and A4 prove the statements in the text connecting the RN measure in Section I to the general case in Section II (see after (A1) and Lemma A1(i)). Indeed, (A13) is the precise analogue to Lemma A1(ii), (A14) is the analogue to Lemma A1(iii), and (A16) gives immediately that $\widetilde{\mathbb{E}}_0^{\diamond}[X_j^* \mid R_T] = \widetilde{\mathbb{E}}_0[X_j^* \mid R_T]$, which was the main implication of Lemma A1(i) used in deriving the results in Section I. We can thus apply those results in this case using the two lemmas above, as follows.

PROOF OF PROPOSITION 3: No arbitrage gives the existence of a positive SDF for which (14) and Assumption 3 are valid. We have

$$\widetilde{\pi}_{t,j} = \mathbb{E}_t[\widetilde{\pi}_{t+1,j}], \qquad \widetilde{\pi}_{t,j}^* = \widetilde{\mathbb{E}}_t^{\diamond}[\widetilde{\pi}_{t+1,j}^*],$$

$$\widetilde{\mathbb{E}}_0^{\diamond}[X_j^*] = 0, \qquad \widetilde{\mathbb{E}}_0^{\diamond}[X_j^* \mid R_T] = \widetilde{\mathbb{E}}_0[X_j^* \mid R_T],$$

where the first equality uses LIE and the rest use Lemmas A3 and A4. The last equation implies, using the argument applied in Lemma A2, that $\triangle_j \leqslant \widetilde{\pi}_{0,j}^*$. Further, (5) to (8) hold for $\widetilde{\pi}_{t,j}$, $\widetilde{\pi}_{t,j}^*$, ϕ_j. We have thus obtained all the conditions used for Lemma 1, Propositions 1 and 2, and Corollary 1, and thus those results continue to hold, with $\widetilde{\pi}_{t,j}^*$ replacing π_t^*, $\widetilde{\pi}_{t,j}$ replacing π_t, X_j^* replacing X^*, ϕ_j replacing ϕ, $\widetilde{\mathbb{E}}_0[\cdot]$ replacing $\mathbb{E}[\cdot]$, and $\triangle_j \equiv \widetilde{\mathbb{E}}_0[X_j^* \mid R_T = \theta_{j+1}] - \widetilde{\mathbb{E}}_0[X_j^* \mid R_T = \theta_j]$ replacing $\triangle$, as stated. □

PROOF OF PROPOSITION 4: The result follows immediately from equation (7), with S_j and S_{j+1} replacing $C_{T,1}$ and $C_{T,0}$, respectively. □

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.